1 Introduction

The overarching aim of the Local Langlands Correspondence (LLC) is to classify the smooth admissible complex representations of reductive p-adic groups in terms of (hopefully simpler) geometric and number theoretic data. More concretely, if G is a reductive p-adic group over some non-archimedean field F, the Langlands philosophy predicts that the (smooth admissible) complex representations should be controlled by the Weil group $\mathcal{W}_F$ of the field F, together with the geometry of the complex dual group $G^\vee$ of G. In this paper, we assume throughout that G is a simple, split p-adic group of simply connected type. Our main results assume, in addition, that G is of exceptional type, but this is not relevant for the purposes of this introduction. Under these assumptions, G is uniquely determined by the p-adic field F and the type of the underlying root system of G. Moreover, the dual group $G^\vee$ is a simple complex group of adjoint type. In this setting, LLC predicts the existence of a bijection between

$$\mathrm{LLC}: \left\{\begin{array}{c}\text{Irreducible smooth}\\ \text{admissible complex}\\ \text{reps of } G(F)\end{array}\right\} \twoheadrightarrow \left\{\begin{array}{c}\text{Langlands parameters}\\ \varphi: \mathcal{W}_F \times \mathrm{SL}_2(\mathbb{C}) \to G^\vee(\mathbb{C})\end{array}\right\},$$

satisfying certain natural compatibility conditions, where the first set is taken up to isomorphism and the second one up to conjugation by $G^\vee(\mathbb{C})$. The fibres of this map are called L-packets and they are a central object of interest in the LLC. By including additional geometric data, it is possible to distinguish between distinct elements in an L-packet. This is achieved by refining Langlands parameters: in addition to the map $\varphi : \mathrm{SL}_2(\mathbb{C}) \times \mathcal{W}_F \to G^\vee(\mathbb{C})$ one considers representations ϕ of the component group A_φ of the centralizer $Z_{G^\vee}(\mathrm{Im}(\varphi))$ of the image of φ in $G^\vee$. These representations $\phi \in \mathrm{Irr}(A_\varphi)$ parametrize the distinct elements in the L-packet corresponding to φ.

Over the past few decades, the local Langlands correspondence has sparked remarkable interest, precise conjectures have been stated and many important results are now known. While the arithmetic of F is important to classify all representation of the p-adic group G (visible through the Weil group $\mathcal{W}_F$), the geometry of the complex dual group $G^\vee$ is enough to classify the class of *unipotent representations* of G. To define this class of representations, one considers first unramified Langlands parameters trivial on the inertia subgroup $\mathcal{I}_F$ of the field F. By the Jacobson–Morozov theorem, unramified Langlands parameters φ are determined, up to the action of $G^\vee$, by the commuting unipotent and semisimple elements $u := \varphi(1, \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right))$ and $s := \varphi(\mathrm{Frob}, \mathrm{Id})$. Moreover, the component group A_φ becomes the component group A_{us} of the centralizer $Z_{G^\vee}(us)$ in $G^\vee$. The set of unramified Langlands parameters up to $G^\vee$-conjugation is therefore in bijection with the set $\{(x,\phi), x \in G^\vee, \phi \in \widehat{A_x}\}$ up to $G^\vee$-conjugation. The set of unipotent representations of G, denoted by $\mathrm{Irr}_{\mathrm{un}}(G)$, fit in the commutative diagram

$$\begin{tikzcd}
\left\{ \begin{array}{c} \text{Unipotent} \\ \text{representations of } G \end{array} \right\}_{/\cong} \arrow[r, "\cong"] \arrow[d, hook] & \{(x,\phi) : x \in G^\vee, \phi \in \mathrm{Irr}(A_x)\}_{/G^\vee} \arrow[d, hook] \\
\left\{ \begin{array}{c} \text{Irreducible smooth} \\ \text{admissible complex} \\ \text{reps of } G(F) \end{array} \right\}_{/\cong} \arrow[r, "\cong"] & \left\{ \begin{array}{c} \text{Pairs } (\varphi,\phi) : \\ \varphi : \mathrm{SL}_2(\mathbb{C}) \times \mathcal{W}_F \to G^\vee(\mathbb{C}) \\ \text{and } \phi \in \mathrm{Irr}(A_\varphi) \end{array} \right\}_{/G^\vee}
\end{tikzcd}$$

Figure 1: Langlands parametrization of unipotent representations of G.

We denote by $\pi(x, \phi)$ the unipotent representation corresponding to the Langlands parameter (x, ϕ). The class of unipotent representations contains all representations containing a vector fixed by an Iwahori subgroup of G (usually denoted *Iwahori-spherical* representations). This was first realized by the work of Kazhdan and Lusztig in [KL87], where they classified Iwahori-spherical representations in terms of geometric data related $G^\vee$. They proved that it was possible to associate, to each Iwahori-spherical representation, a unique pair (x, ϕ) up to $G^\vee$-conjugacy. This association is generally not surjective; however, each representation $\pi(x, \mathbf{1}), x \in G^\vee$ is Iwahori-spherical. Consequently, unipotent representations are the union of all L-packets containing some Iwahori-spherical representation.

Lusztig's subsequent work showed that Deligne–Lusztig theory can be used to define unipotent representations of G explicitly in terms of the p-adic topology and structure of G alone. We devote Section 2 to discuss this construction in some detail. The Langlands parametrization of unipotent representations of G as defined by Lusztig in terms of geometric data (see Figure 1) has come from many years of development. Lusztig's work [Lus95], [Lus02] settled it for adjoint p-adic groups, Reeder [Ree02] for Iwahori-spherical representations of split groups of arbitrary isogeny, and Solleveld for all reductive p-adic groups [Sol23a],[Sol23b].

While many questions in the unipotent LLC have now been resolved in full generality, some of them remain open. One such question relates to the stability of L-packets. Moeglin and Waldspurger tackled this problem for $G = \mathrm{SO}_{2n+1}(F)$ in [MW03] by considering the restriction of unipotent representations to maximal open compact subgroups. These subgroups provide a passage from unipotent representations of the p-adic group to unipotent representations of certain finite reductive groups. Work of Lusztig and others in the late 20th century revealed remarkable structure on unipotent characters of these finite reductive groups, which Moeglin and Waldspurger exploited. Lusztig defined an involution on the space of virtual unipotent characters known as *Lusztig's non-abelian transform* taking the irreducible characters to the so-called *almost characters*. These characters can be obtained independently via the theory of character sheaves and, as a result, have nice geometrical and stability properties.

The hope is that, if we can lift the Fourier transform to the setting of p-adic groups with analogous properties, one could use the resulting virtual characters to prove stability properties of L-packets.

Moreover, it is believed by Lusztig that an analogous theory of character sheaves for p-adic groups should exist. Work of Ciubotaru [Ciu20] and Aubert–Ciubotaru–Romano [ACR25] has already considered this problem and formulated precise conjectures. We consider the same setting and similar notation to [ACR25], which we now describe in the simplified setting when G is of simply connected type. Let $\max(G)$ denote the set of conjugacy classes of maximal open compact subgroups of G, which coincides with the set of conjugacy classes of maximal parahoric subgroups. Each $K \in \max(G)$ has a pro-p unipotent radical U_K with reductive quotient $\overline{K}$. When G is simply connected, $\overline{K}$ is a connected finite group of Lie type. The *unipotent parahoric restriction map* is the linear transformation

$$\begin{aligned} \mathrm{res}_{\mathrm{un}}^{K} : R_{\mathrm{un}}(G) &\longrightarrow R_{\mathrm{un}}(\overline{K}) \\ V &\longmapsto V^{U_K}, \end{aligned}$$

where $R_{\mathrm{un}}(G)$ and $R_{\mathrm{un}}(\overline{K})$ are the complexification of the Grothendieck groups spanned by the unipotent representations of G and $\overline{K}$, respectively. Secondly, Lusztig's nonabelian Fourier transforms on $R_{\mathrm{un}}(\overline{K})$ for each $K \in \max(G)$ give an involution

$$\mathrm{FT}_{\mathrm{un}}^{\mathrm{cpt}} : \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}) \longrightarrow \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}).$$

The remaining ingredient to formally state Aubert–Ciubotaru–Romano's conjecture [ACR25, Conjecture 1.2] is the construction of the *dual nonabelian Fourier transform* $\mathrm{FT}^{\vee} : R_{\mathrm{un}}(G) \longrightarrow R_{\mathrm{un}}(G)$ compatible with the unipotent restriction maps $\mathrm{res}_{\mathrm{un}}^{K}$ and Lusztig's nonabelian Fourier transform $\mathrm{FT}^{\overline{K}}$. The existence of such a map is conjectured, but so far no explicit general construction has been achieved on the infinite dimensional space $R_{\mathrm{un}}(G)$. Instead, Ciubotaru–Opdam [CO17] realized that, for applications, it was often sufficient to study the space quotient space $\overline{R}_{\mathrm{un}}(G)$ of $R_{\mathrm{un}}(G)$ by the span of all properly parabolically induced representations, since parabolic induction is generally well understood. The upshot of this simplification is double: firstly, the quotient space $\overline{R}_{\mathrm{un}}(G)$ is finite dimensional, in contrast to the infinite dimensional space $R_{\mathrm{un}}(G)$. Secondly, $\overline{R}_{\mathrm{un}}(G)$ is canonically isomorphic to the elliptic unipotent representation space $R_{\mathrm{un,ell}}(G)$, a certain subspace of $R_{\mathrm{un}}(G)$. Moreover, the Local Langlands correspondence provides an explicit description for a natural orthogonal basis of $R_{\mathrm{un,ell}}(G)$. To describe this basis, let $u \in G^{\vee}$ be a unipotent element and let Γ_u be the reductive part of the centralizer $Z_{G^{\vee}}(u)$. We then consider the set of elliptic pairs

$$\mathcal{Y}(\Gamma_u)_{\mathrm{ell}} = \{(s,h) \in \Gamma_u \mid sh = hs \text{ and } Z_{\Gamma_u}(s,h) \text{ finite}\}.$$

The Local Langlands correspondence induces an isomorphism

$$\mathrm{LLC}_{\mathrm{un}}^{\mathrm{ell}} : \bigoplus_{u \in G^{\vee}} \mathbb{C}[\mathcal{Y}(\Gamma_u)]^{\Gamma_u} \longrightarrow R_{\mathrm{un,ell}}(G), \qquad (u,s) \longmapsto \Pi(u,s,h),$$

where the sum in the left hand side is taken over conjugacy classes of unipotent elements in $G^{\vee}$ and the virtual characters are defined by

$$\Pi(u,s,h) := \sum_{\phi\in A_{us}} \phi(h)\pi(us,\phi).$$

The space of elliptic pairs has an obvious involution given by the flip $(s,h)\mapsto(h,s)$, and this defines an involution

$$\mathrm{FT}^{\vee}_{\mathrm{ell}} : R_{\mathrm{un,ell}}(G) \longrightarrow R_{\mathrm{un,ell}}(G),$$

the dual elliptic nonabelian Fourier transform. We are finally ready to state the main conjecture of this paper, in the simplified setting of split, simply connected p-adic groups.

Conjecture 1.1. [ACR25, Conjecture 1.2] Let G be a split semisimple p-adic group of simply connected type. Then the diagram

$$\begin{tikzcd}
R_{\mathrm{un,ell}}(G) \arrow[r, "\mathrm{FT}^{\vee}_{\mathrm{ell}}"] \arrow[d, "\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}"] & R_{\mathrm{un,ell}}(G) \arrow[d, "\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}"] \\
\bigoplus_{K\in\max(G)} R_{\mathrm{un}}(\overline{K}) \arrow[r, "\mathrm{FT}^{\mathrm{cpt}}_{\mathrm{un}}"] & \bigoplus_{K\in\max(G)} R_{\mathrm{un}}(\overline{K}),
\end{tikzcd}$$

commutes, up to certain roots of unity.

In their paper, Aubert, Ciubotaru and Romano showed that this conjecture holds for $G = \mathrm{SL}_n(F)$, $\mathrm{PGL}_n(F)$ or $\mathrm{Sp}_4(F)$. In previous work, Ciubotaru and Opdam [Ciu20] showed it holds for $G_2(F)$, and Waldspurger [Wal18] showed it for $G = \mathrm{SO}_{2n+1}(F)$. In this document we settle the above conjecture for two further exceptional cases.

Theorem 1.2. *Let G be a simple, simply connected p-adic group of type F_4 or E_6. Then conjecture 1.1 holds for G.*

Structure of the document

The document is organized as follows. We devote section 2 to introduce finite groups of Lie type G^F and Deligne–Lusztig theory. There is a vast amount of theory, so we focus on understanding the classification of unipotent characters of G^F in terms of families. This naturally leads to the construction of Lusztig's nonabelian Fourier transform. In section 3 we explain relevant structural and representation theoretic results of reductive p-adic groups G, where we introduce the Bruhat-Tits building, unipotent representations and the dual nonabelian Fourier transform. Section 4 is the most technical; there we discuss in detail how to explicitly compute the restriction functors $\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}$, a method developed by Reeder in [Ree00]. Sections 5 and 6 contain original work and prove rewrite this bit.

2 Unipotent representations for finite groups of Lie type

Despite the fact that the statement of Conjecture 1.1 is at heart a result about the representation theory of a p-adic group G, in this first section we briefly recall important representation theoretic properties of finite groups of Lie type leading to the Lusztig's construction of the nonabelian fourier transform described in the introduction. Suppose $\mathbb{G}$ is a connected reductive group over $\overline{\mathbb{F}_p}$ with Frobenius map $\mathrm{Fr} : \mathbb{G} \to \mathbb{G}$. The group $\mathbb{G}^{\mathrm{Fr}}$ is a finite group inheriting many structural properties of $\mathbb{G}$. An easy application of the Lang–Steinberg theorem [Car85, §1.17] gives the existence of Fr-stable Borel subgroups of $\mathbb{G}$, and each Fr-stable Borel contains an Fr-stable maximal torus of $\mathbb{G}$. While all Fr-stable Borel subgroups of $\mathbb{G}$ are conjugate under $\mathbb{G}^{\mathrm{Fr}}$, this is not the case for Fr-stable maximal tori. The $\mathbb{G}^{\mathrm{Fr}}$-conjugacy classes of Fr-stable maximal tori are classified by the Fr-conjugacy classes of the Weyl group $W := N_{\mathbb{G}}(\mathbb{T})/\mathbb{T}$, where $\mathbb{T}$ is any Fr-stable maximal tori of $\mathbb{G}$.

The Frobenius map Fr induces an action σ on the Dynkin diagram by $\mathrm{Fr}(\mathcal{X}_\alpha) = \mathcal{X}_{\sigma(\alpha)}$, where the simple roots are chosen appropriately. When this action σ is trivial, we say that $\mathbb{G}^{\mathrm{Fr}}$ is split, in which case the $\mathbb{G}^{\mathrm{Fr}}$-conjugacy classes of Fr-maximal tori are classified by the W-conjugacy classes. All finite groups of Lie type considered in this paper are split, so we assume hereafter that Fr induces the trivial automorphism of the Dynkin diagram. In this case, conjugacy classes of the Weyl groups parametrize classes of Fr-stable tori. Given $w \in W$, we denote by $\mathbb{T}_w \subseteq \mathbb{G}$ an Fr-maximal torus in $\mathbb{G}^{\mathrm{Fr}}$-conjugacy class parametrized by w. For example, the class of $\mathbb{T}_e$ contains all Fr-stable maximal tori contained in Fr-stable Borel subgroups. Deligne and Lusztig's groundbreaking work associates, to each Fr-maximal torus $\mathbb{T}$ and character $\theta : \mathbb{T}^{\mathrm{Fr}} \to \mathbb{C}^\times$ a virtual character $R_{T,\theta}$ of the group $\mathbb{G}^{\mathrm{Fr}}$ with striking properties. This construction is widely known as *Deligne–Lusztig induction*. The original paper of Deligne and Lusztig [DL76] or Carter's book [Car85, §7] are great references that discuss in detail many of these properties.

Using the virtual characters $R_{\mathbb{T},\theta}$, Deligne and Lusztig defined a representation $\rho \in \mathrm{Irr}(\mathbb{G}^{\mathrm{Fr}})$ to be *unipotent* if $\langle R_{\mathbb{T},1}, \rho\rangle \neq 0$ for some Fr-stable maximal torus $\mathbb{F}$ of $\mathbb{G}$. In particular, this implies that ρ arises as a representation of $\mathbb{G}^{\mathrm{Fr}}$ from an ℓ-adic cohomology space of some Deligne–Lusztig variety. We remark that $R_{T_e,1} = \mathrm{Ind}_{\mathbb{B}^{\mathrm{Fr}}}^{\mathbb{G}^{\mathrm{Fr}}} 1$, so unipotent representations naturally extend principal series representations. In addition, unipotent representations naturally factor through the centre of $\mathbb{G}^{\mathrm{Fr}}$, so one normally assumes without loss of generality that $\mathbb{G}$ is of adjoint type. We denote the set of unipotent characters by $\mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$.

The irreducible characters ϕ of W are known to parametrize principal series representations χ_ϕ of $\mathbb{G}^{\mathrm{Fr}}$. Lusztig considered in addition the virtual character

$$R_\phi := \frac{1}{|W|} \sum_{w \in W} \phi(w) R_{\mathbb{T}_w,1}$$

which he called an *almost character* of $\mathbb{G}^{\mathrm{Fr}}$. These characters arise as traces of character sheaves of $\mathbb{G}$

and thus have nice geometric stability properties. These characters are a natural orthonormal basis of $R_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$, the complexification of the Grothendieck space spanned by the unipotent characters of $\mathbb{G}$. The above construction gives a canonical map

$$\{\text{principal series reps}\} \longrightarrow \{\text{almost characters}\}, \qquad \chi_\phi \longmapsto R_\phi.$$

Lusztig realized that it was possible to naturally extend the above map for all unipotent representations obtaining an involution on $\mathrm{FT}^{\mathbb{G}}$. To achieve this, first we group unipotent representations into families by stating that two representations ρ_1, ρ_2 are in the same family if

$$\langle \rho_1, R_\phi \rangle \neq 0 \quad \text{and} \quad \langle \rho_2, R_\phi \rangle \neq 0 \quad \text{for some } \phi \in \mathrm{Irr}(W),$$

and then extending by transitivity. To each family $\mathcal{U} \subset \mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})$, Lusztig associated a finite group $\Gamma_{\mathcal{U}}$, scalars $\Delta(x,\sigma) \in \{\pm 1\}, (x,\sigma) \in \mathcal{M}(\Gamma_{\mathcal{U}})$ defined in [Lus84] and a bijection

$$\begin{aligned} \mathcal{U} &\longleftrightarrow \mathcal{M}(\Gamma_{\mathcal{U}}) := \{(x,\sigma) \mid x \in \Gamma_{\mathcal{U}}, \sigma \in \mathrm{Irr}(Z_{\Gamma_{\mathcal{U}}}(x))\}/\Gamma_{\mathcal{U}} \\ \rho_{(x,\sigma)} &\longleftrightarrow (x,\sigma). \end{aligned}$$

The indexing set $\mathcal{M}(\Gamma_{\mathcal{U}})$ carries a natural pairing

$$\{(x,\sigma),(y,\tau)\} = \frac{1}{|Z_{\Gamma_{\mathcal{U}}}(x)||Z_{\Gamma_{\mathcal{U}}}(y)|} \sum_{\substack{g \in \Gamma_{\mathcal{U}} \\ xgyg^{-1} = gygx^{-1}}} \sigma(gyg^{-1})\tau(g^{-1}x^{-1}g),$$

and the above bijection satisfies the property that

$$\langle \rho_{(x,\rho)}, R_\phi \rangle_{\mathbb{G}^{\mathrm{Fr}}} = \begin{cases} \Delta(x,\sigma)\,\{(x,\rho),(y,\tau)\} & \text{if } \chi_\phi = \rho_{(y,\tau)} \in \mathcal{U}, \\ 0 & \text{if } \chi_\phi \notin \mathcal{U}. \end{cases} \tag{1}$$

We recall that if W is an irreducible Weyl group, then $\Delta(x,\rho) = 1$ except for the families containing $\phi_{512,11}$ in E_7, $\phi_{4096,11}$ or $\phi_{4096,26}$ in E_8. For any family $\mathcal{U}$, $\rho_{(e,1)} = \chi_\phi$ is a principal series representation corresponding to some representation ϕ of W. Such representations of W are called *special*, and they are the only ones satisfying the property that $\langle \chi, R_\phi \rangle \geq 0$ for all unipotent characters χ (see [Car85, §12.3]). We define the *unipotent almost characters* of $\mathbb{G}^{\mathrm{Fr}}$ to be the set of orthonormal class functions

$$R_{(x,\sigma)} := \sum_{(y,\tau)} \Delta(y,\tau)\{(y,\tau),(x,\sigma)\}\rho_{(y,\tau)}, \quad (x,\sigma) \in \mathcal{M}(\Gamma_{\mathcal{U}}).$$

In particular, if $\chi_\phi = \rho_{(x,\rho)}$, then $R_\phi = R_{(x,\rho)}$. Lusztig also defined the map

$$\begin{aligned}\mathrm{FT}_{\Gamma_\mathcal{U}} : \mathbb{C}[\mathcal{U}] &\longrightarrow \mathbb{C}[\mathcal{U}]\\ \rho_{(x,\sigma)} &\longmapsto \sum_{(y,\tau)\in\mathcal{M}(\Gamma_\mathcal{U})} \{(x,\sigma),(y,\tau)\}\rho_{(y,\tau)}\end{aligned}$$

The *unipotent nonabelian Fourier transform* of $\mathbb{G}^{\mathrm{Fr}}$ on the space $R_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}}) = \oplus_{\mathcal{U}\subseteq \mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})}\mathbb{C}[\mathcal{U}]$ is the map

$$\mathrm{FT}^{\mathbb{G}^{\mathrm{Fr}}} := \bigoplus_{\mathcal{U}\subset \mathrm{Irr}_{\mathrm{un}}(\mathbb{G}^{\mathrm{Fr}})} \mathrm{FT}_{\Gamma_\mathcal{U}},$$

satisfying many properties. Most importantly, it is a self-Hermitian involution with respect to the standard inner product of characters satisfying $\mathrm{FT}^{\mathbb{G}^{\mathrm{Fr}}}(\chi_\phi) = R_\phi$; that is, it takes unipotent representations to the almost characters arising as traces of character sheaves on $\mathbb{G}$.

When $\mathbb{G}^{\mathrm{Fr}}$ is split (as it is the case throughout this document) it is convenient to express Lusztig's nonabelian Fourier transform in term of stable combinations of unipotent representations. This makes the verification of Conjecture 1.1 cleaner and more straightforward. Let $\mathcal{U}$ be a family of unipotent representations and for each $x \in \Gamma_\mathcal{U}$, define the virtual combinations of unipotent characters

$$\Pi_\mathcal{U}(x,y) = \sum_{\sigma\in \widehat{Z_{\Gamma_\mathcal{U}}(x)}} \sigma(y^{-1})\rho_{(x,\sigma)}, \quad \text{where } y \in Z_{\Gamma_\mathcal{U}}(x). \tag{2}$$

Dually, when $\Gamma_\mathcal{U}$ is abelian, define the virtual combinations

$$\Pi_\mathcal{U}(\sigma,\tau) = \sum_{y\in\Gamma_\mathcal{U}} \tau(y)\rho_{(y,\sigma)}, \quad \text{if } \sigma,\tau \in \widehat{\Gamma_\mathcal{U}}. \tag{3}$$

Lemma 2.1 ([ACR25, Lemma 5.1])**.** *With the notation of* (2) *and* (3)*,*

$$\mathrm{FT}_{\Gamma_\mathcal{U}}(\Pi_\mathcal{U}(x,y)) = \Delta(x,y)\Pi_\mathcal{U}(y,x), \quad \mathrm{FT}_{\Gamma_\mathcal{U}}(\Pi_\mathcal{U}(\sigma,\tau)) = \Pi_\mathcal{U}(\tau,\sigma),$$

the latter when $\Gamma_\mathcal{U}$ is abelian, where $\Delta(x,y) = 1$ unless $\mathcal{U}$ is one of the families described in (1)*, in which case $\Delta(1,g_2) = \Delta(g_2,1) = -1$ and 1 otherwise.*

3 Unipotent representations of p-adic groups

In this section, we bridge the gap between the representation theory of finite groups of Lie type and that of p-adic groups briefly discussed during the introduction. We preserve the same setting: throughout, we let G be a connected simple group defined over a finite extension F of $\mathbb{Q}_p$ with residue field $\mathbb{F}_q$. We further assume that G is split with split maximal torus T and of simply connected type. In particular, G is determined by the p-adic field F and the type of the root system Φ of G. Let $\mathrm{Irr}(G)$ be the set of irreducible smooth representations of G and $\mathrm{Irr}_{\mathrm{un}}(G) \subseteq \mathrm{Irr}(G)$ the subset of unipotent representations. We now describe Lusztig's characterization of unipotent representations of G in terms of the topology G and the representation theory of finite groups of Lie type.

We associate the Bruhat–Tits building $\mathcal{B}(G)$ to the group G, a simplicial complex whose dimension agrees with the rank of G and carrying an action of G by simplicial transformations. To each facet c in the building of G, consider the *parahoric subgroup* $K_c := \mathrm{Stab}_G(c)$. When G is simply connected, K_c is a connected open compact subgroup of G fitting in an exact sequence

$$1 \longrightarrow U_c \longrightarrow K_c \longrightarrow \overline{K}_c \longrightarrow 1,$$

where U_c is the pro-unipotent radical of K_c and $\overline{K}_c$ is the group of $\mathbb{F}_q$-rational points of a connected reductive group $\overline{\mathbf{K}}_{\mathbf{c}}$. Since $K_{g\cdot c} = gK_cg^{-1}$ for all facets c $g \in G$, conjugacy classes of parahoric subgroups are in bijection with G-orbits of facets in $\mathcal{B}(G)$. Furthermore, we recall that after fixing a set of simple affine roots S_{aff} of G with corresponding fundamental alcove $\mathcal{C}_0$, there is a bijection

$$\begin{aligned}\{\text{Subsets } J \subsetneq S_{\mathrm{aff}}\} &\longleftrightarrow \{G\text{- orbits of facets in } \mathcal{B}(G)\}\\ J &\longmapsto c_J = \{x \in \overline{\mathcal{C}_0} \mid \langle \alpha, x\rangle \in \mathbb{Z} \iff \alpha \in J, \text{ for any } \alpha \in S_{\mathrm{aff}}\}\end{aligned}$$

satisfying that $J_1 \subseteq J_2$ if and only if $c_{J_2} \subset \overline{c_{J_1}}$. Throughout, we denote by $(K_J, U_J, \overline{K}_J)$ the parahoric subgroup corresponding to the facet c_J. In particular, maximal parahoric subgroups corresponds to facets of minimal dimension and maximal subsets $J \subsetneq S_{\mathrm{aff}}$. When G is simply connected, we recall that the set of maximal parahoric subgroups coincides with the set of maximal open compact groups. We define

$$\max(G) := \{K_x \mid \{x\} \subset \mathcal{B}(G) \text{ facet of minimal dimenson}\}$$

to be the set containing each orbit of maximal parahoric subgroups.

Next, let (π, V) a smooth admissible complex representation of G and let $(K, U, \overline{K})$ be a parahoric subgroup. The space V^{U_K} of fixed points under the pro-unipotent radical is naturally a representation of $\overline{K} = K/U_K$. This induces the *parahoric restriction functor*

$$\mathrm{res}_K : R(G) \longrightarrow R(\overline{K}), \quad V \longmapsto V^{U_K}, \quad \text{for all } V \in \mathrm{Irr}(G), \tag{4}$$

where $R(G)$ and $R(\overline{K})$ are the complexification of the Grothendieck group of $\mathrm{Irr}(G)$ and $\mathrm{Irr}(\overline{K})$. If (σ, E) is a cuspidal representation of $\overline{K}$, we define

$$\mathrm{Irr}(G, K, E) = \{(\pi, V) \in \mathrm{Irr}(G) \mid \text{ the } \overline{K}\text{-module contains the } \overline{K}\text{-module } E\}.$$

Definition 3.1. We say that an irreducible representation (π, V) of G is *unipotent* if there is a parahoric subgroup $(K, U_K, \overline{K})$ such that V^{U_K} contains a cuspidal unipotent representation of $\overline{K}$; that is, if $(\pi, V) \in \mathrm{Irr}(G, K, E)$ for some pair (K, E) where E is unipotent.

Therefore, the set of unipotent representations up to isomorphism is

$$\mathrm{Irr}_{\mathrm{un}}(G) = \bigcup_{\substack{J \subsetneq S_{\mathrm{aff}} \\ E \text{ cusp. unip. } \overline{K}_J\text{-rep}}} \mathrm{Irr}(G, K_J, E).$$

Each set $\mathrm{Irr}(G, K, E)$ generates a distinct Bernstein *block* of representations of G. If $\mathcal{I}$ is an Iwahori subgroup, $\mathcal{I}/U_{\mathcal{I}} \cong \mathbb{F}_q^\times$ and the only unipotent is the trivial character. Thus, Iwahori-spherical representations are all unipotent. Moreover, unipotent representations behave well under the parahoric restriction functors res^K. The first important result is due to Moy and Prasad. A similar result holds for non-unipotent representations, but it is more complicated.

Proposition 3.2. [Ree00, Lemma 5.1] *Assume G is simply connected. Suppose $I \subsetneq S_{\mathrm{aff}}$ and that V^{U_I} contains a cuspidal unipotent representation σ of $\overline{K}_I$. If $J \subsetneq S_{\mathrm{aff}}$ with $V^{U_J} \neq 0$, and J is minimal with respect to this property $I = J$.*

The following is a direct consequence of the above result.

Corollary 3.3. *For any two pairs (K, E) and (K', E') of a parahoric subgroup and a cuspidal unipotent representation of the reductive quotient,* $\mathrm{Irr}(G, K, E)$ *and* $\mathrm{Irr}(G, K', E')$ *are either disjoint or equal.*

Proposition 3.2 together with a standard Harish–Chandra style argument yields the following corollary.

Corollary 3.4. *Let (π, V) be a unipotent representation of G and let $(K, U_K, \overline{K})$ be a parahoric subgroup. Then the $\overline{K}$-irreducible components of V^{U_K} are all unipotent.*

Consequently, for any parahoric subgroup K, the parahoric restriction functor res^K naturally restricts to the *unipotent parahoric restriction function*

$$\mathrm{res}^K_{\mathrm{un}} : R_{\mathrm{un}}(G) \longrightarrow R_{\mathrm{un}}(\overline{K}), \quad V \longmapsto V^{U_K},$$

where $R_{\mathrm{un}}(G)$ and $R_{\mathrm{un}}(\overline{K})$ are the complexification of the Grothendieck groups spanned by $\mathrm{Irr}_{\mathrm{un}}(G)$ and $\mathrm{Irr}_{\mathrm{un}}(\overline{K})$, respectively. Finally, we let

$$\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}} := \bigoplus_{K \in \max(G)} \mathrm{res}^K_{\mathrm{un}} : R_{\mathrm{un}}(G) \longrightarrow \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}).$$

3.1 The Langlands classification and the dual nonabelian Fourier transform

The Local Langlands provides a beautiful classification of the unipotent representations of G in terms of the geometry of the dual group $G^\vee$ of G, of adjoint type. This result is widely known as Lusztig's arithmetic-geometric correspondence.

Theorem 3.5 (The arithmetic-geometric correspondence [Lus95])**.** *There is a natural bijection between*

$$G^\vee \backslash \{(x,\phi) \mid x \in G^\vee, \phi \in \widehat{A_x}\} \longleftrightarrow \mathrm{Irr}_{\mathrm{un}}(G)$$
$$(x,\phi) \longmapsto (\pi_{(x,\phi)}, V_{(x,\phi)}).$$

In particular, there is an isomorphism

$$\mathrm{LLC}_{\mathrm{un}} : \bigoplus_{x\in G^\vee} R(A_x) \longrightarrow R_{\mathrm{un}}(G), \tag{5}$$

where the sum on the left hand side is taken over conjugacy classes of $G^\vee$ and $R(A_x)$ is the space of virtual characters of A_x.

Given a Langlands parameter (x,ϕ), it is a natural question to ask inside which unipotent block does the representation $(\pi_{(x,\phi)}, V_{(x,\phi)})$ lies inside of. This is the content of the generalized Springer correspondence, which we will not revise in detail here (see [CMBO25, §4.3]) Instead, we will state a fundamental result giving a geometric characterization for Iwahori-spherical representations in terms of the data (x,ϕ). These representations lie in the *principal block*, generated by $\mathrm{Irr}(G,\mathcal{I},\mathbf{1})$. For the other representations that are not Iwahori-spherical, we will use Reeder's computations in [Ree00, §10,11]. To state this result, we note that given the data (x,ϕ), one can consider the partial flag variety $\mathcal{B}^x$ containing all Borel subgroups of $G^\vee$. This variety carries a natural action of A_x given by conjugation. In particular, the cohomology complex $H^*(\mathcal{B}^x)$ inherits an action of A_x too.

Theorem 3.6. [KL87, Theorem 7.12] *The representation $(\pi_{(x,\phi)}, V_{(x,\phi)})$ is Iwahori-spherical if and only if $H^*(\mathcal{B}^x)^\phi \neq 0$. In particular, for any $x \in G^\vee$, the representation $(\pi_{(x,\mathbf{1})}, V_{(x,\mathbf{1})})$ is Iwahori-spherical.*

In this section we also define the dual nonabelian Fourier transform for unipotent representations of p-adic groups. Conjecturally, we expect the existence of a map $\mathrm{FT}^\vee$, a dual nonabelian Fourier transform, on the unipotent representation space $R_{\mathrm{un}}(G)$ with analogous properties to Lusztig's nonabelian Fourier transforms $\mathrm{FT}^{\overline{K}}$ and compatible with the unipotent parahoric restriction maps $\mathrm{res}^K_{\mathrm{un}}$. However, such a map has not been constructed in the literature in general on the infinite dimensional space $R_{\mathrm{un}}(G)$. For some applications, it is enough to construct it on the elliptic unipotent representation space $R_{\mathrm{un,ell}}(G)$, a finite dimensional subspace of $R_{\mathrm{un}}(G)$ which we now describe.

Classically, the elliptic unipotent representation space $\overline{R}_{\mathrm{un}}(G)$ is defined as the quotient of $R_{\mathrm{un}}(G)$ by the subspace spanned bya ll properly parabolically induced representations. Equivalently, $\overline{R}_{\mathrm{un}}(G)$ is obtained by computing the quotient of $R_{\mathrm{un}}(G)$ by the radical of the Euler–Poincare pairing

$$EP_G(\ ,\ ): R_{\mathrm{un}}(G) \times R_{\mathrm{un}}(G) \longrightarrow \mathbb{C}, \quad (V, V') \longmapsto \sum_{i \geq 0} (-1)^i \mathrm{Ext}^i(V, V'), \text{ where } V, V' \in \mathrm{Irr}_{\mathrm{un}}(G),$$

and $EP_G(\ ,\ )$ is obtained by linearity, and where $\mathrm{Ext}^i(V, V')$ are calculated in the category of smooth complex G-representations. Thus, the space $R_{\mathrm{un}}(G)$ carries a natural non-degenerate pairing $EP_G(\ ,\ )$. Similarly, for each $x \in G^\vee$ the space of virtual representations $R(A_x)$ carries a natural elliptic pairing $(\ ,\ )^{A_x}_{\mathrm{ell}}$ (see [ACR25, §8.1] for the definition), and $\overline{R}(A_x)$ is the quotient of $R(A_x)$ by the radical of the elliptic pairing. The arithmetic-geometric (5) induces an isometric isomorphism on the elliptic representation spaces

$$\overline{\mathrm{LLC}}_{\mathrm{un}} : \bigoplus_{x \in G^\vee} \overline{R}(A_x) \longrightarrow \overline{R}_{\mathrm{un}}(G), \tag{6}$$

where the sum on the left is again over conjugacy classes of elements in $G^\vee$. The left hand side of (6) can be further decomposed into

$$\bigoplus_{x \in G^\vee} \overline{R}(A_x) = \bigoplus_{\substack{u \in G^\vee \\ \text{unipotent}}} \bigoplus_{\substack{s \in Z_{G^\vee}(u) \\ \text{semisimple}}} \overline{R}(A_{su}).$$

Next, fix some unipotent element $u \in G^\vee$ and let Γ_u be the reductive part of the centralizer $Z_{G^\vee}(u)$. We define the set of elliptic pairs associated to u to be

$$\mathcal{Y}(\Gamma_u)_{\mathrm{ell}} = \{(s, h) \in \Gamma_u \times \Gamma_u \mid s, h \text{ semisimple}, sh = hs, Z_{\Gamma_u}(s, h) \text{ is finite}\}.$$

It is not hard to show that this set is always finite (see [ACR25, Lemma 8.1]). For every $(s, h) \in \mathcal{Y}(\Gamma_u)$, define

$$\Pi(s, h) = \sum_{\phi \in \widehat{A_{\Gamma_u}(s)}} \phi(\overline{h}) \phi \in R(A_{\Gamma_u}(s)), \tag{7}$$

where $\overline{h}$ is the image of h in $A_{\Gamma_u}(s)$. Moreover, let $\overline{\Pi}(s, h)$ denote the image in $\overline{R}(A_{\Gamma_u}(s))$ and let $\mathbb{C}[\mathcal{Y}(\Gamma_u)_{\mathrm{ell}}]^{\Gamma_u} \cong \mathbb{C}[\Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}]$ be the space of Γ_u-invariant functions $\mathcal{Y}(\Gamma_u)$. Let $\mathbf{1}_{[(s,h)]}$ be the indicator function of the Γ_u-orbit of (s, h).

Proposition 3.7. [ACR25, Proposition 8.3] *Let $u \in G^\vee$ be unipotent. Then the correspondence $\mathbf{1}_{[(s,h)]} \mapsto \overline{\Pi}(s, h)$ induces an isomorphism*

$$\mathbb{C}[\mathcal{Y}(\Gamma_u)_{\mathrm{ell}}]^{\Gamma_u} \longrightarrow \bigoplus_{\substack{s \in Z_{G^\vee}(u) \\ \text{semisimple}}} \overline{R}(A_{su})$$

Finally, for each $[(s,h)] \in \Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}$, define the virtual combinations

$$\Pi(u,s,h) := \sum_{\phi \in \widehat{A_{\Gamma_u}(s)}} \phi(h)\pi(su,\phi) \in R_{\mathrm{un}}(G), \tag{8}$$

and let $\overline{\Pi}(u,s,h)$ be the corresponding images in $\overline{R}_{\mathrm{un}}(G)$.

Definition 3.8. The elliptic unipotent representation space $R_{\mathrm{un,ell}}(G)$ is the finite-dimensional subspace of $R_{\mathrm{un}}(G)$ spanned by virtual characters $\{\Pi(u,s,h) \mid u \text{ unipotent, and } (s,h) \in \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}\}$.

Thus, the isomorphism (6) together with Proposition 3.7 yields an isometric isomorphism

$$R_{\mathrm{un,ell}}(G) \longrightarrow \overline{R}_{\mathrm{un}}(G), \quad \Pi(u,s,h) \mapsto \overline{\Pi}(u,s,h) \tag{9}$$

The set $\Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}$ admits an obvious involution given by flip : $(s,h) \mapsto (h,s)$, which induces an involution on $R_{\mathrm{un,ell}}(G)$.

Definition 3.9. The *dual elliptic nonabelian Fourier transform* is the involutive linear map $\mathrm{FT}^{\vee}_{\mathrm{ell}} : R_{\mathrm{un,ell}} \to R_{\mathrm{un,ell}}$ defined by

$$\mathrm{FT}^{\vee}_{\mathrm{ell}}(\Pi(u,s,h)) = \Pi(u,h,s), \quad u \in G^{\vee} \text{ unipotent, } (s,h) \in \Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}.$$

We are now ready to state the main conjecture for elliptic unipotent representations.

Conjecture 3.10. [ACR25, Conjecture 1.2] Let G be a split semisimple p-adic group of simply connected type. There is a commutative diagram, up to certain roots of unity,

$$\begin{tikzcd} R_{\mathrm{un,ell}}(G) \arrow[r, "\mathrm{FT}^{\vee}_{\mathrm{ell}}"] \arrow[d, "\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}"] & R_{\mathrm{un,ell}}(G) \arrow[d, "\mathrm{res}^{\mathrm{cpt}}_{\mathrm{un}}"] \\ \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}) \arrow[r, "\mathrm{FT}^{\mathrm{cpt}}_{\mathrm{un}}"] & \bigoplus_{K \in \max(G)} R_{\mathrm{un}}(\overline{K}), \end{tikzcd}$$

where $\mathrm{FT}^{\mathrm{cpt}}_{\mathrm{un}} := \bigoplus_{K \in \max(G)} \mathrm{FT}^{\overline{K}}_{\mathrm{un}}$ on each coordinate.

The horizontal maps in the above conjecture are easily computable. They are involutions on finite dimensional representation spaces with canonical basis, and computing the matrices is straightforward. The difficulty of the above conjecture relies on the computation of the unipotent parahoric restriction maps $\mathrm{res}^{K}_{\mathrm{un}}$. This is due to the fact that $R_{\mathrm{un}}(G)$ has a natural basis in terms of the geometry of the dual group $G^{\vee}$, while irreducible characters $R_{\mathrm{un}}(\overline{K})$ are parametrized by families and Lusztig's labels. The connection between these parametrizations is deep, and we devote the next section to discuss this in detail for Iwahori-spherical representations. At the end we also give a brief discussion on the computation of this map for non-Iwahori-spherical representations. We closely follow Reeder's discussion in [Ree00, §8,9].

4 Parahoric restriction of unipotent representations

Let G be a connected split simple group of adjoint type over a p-adic field F. In the previous section we stated the bijection

$$\begin{aligned} \mathrm{LLC}^p_{\mathrm{un}} : G^\vee \backslash \{(x,\phi) \mid x \in G^\vee, \phi \in \widehat{A_x}\} &\longleftrightarrow \bigsqcup_{G' \in \mathrm{InnT}^p(G)} \mathrm{Irr}_{\mathrm{un}}(G') \\ (x,\phi) &\longmapsto (\pi_{(x,\phi)}, V_{(x,\phi)}), \end{aligned}$$

a result that parametrizes unipotent representations of G in terms of the geometry of its complex dual group $G^\vee$ and lies in the heart of the local Langlands correspondence.

Let $J \subsetneq S_{\mathrm{aff}}$ be a proper subset of affine roots and let K_J be the corresponding parahoric subgroup of G. In this section we discuss how to explicitly compute the parahoric restriction maps

$$\mathrm{res}^{K_J}_{\mathrm{un}} : R_{\mathrm{un}}(G) \longrightarrow R_{\mathrm{un}}(\overline{K}_J), \quad V \longmapsto V^{U_J}$$

from unipotent representations of G to unipotent representations of $\overline{K}_J$. To do this, we closely follow the discussion from [Ree00, §4-8], which we recall for convenience. Let us fix throughout some proper subset $J \subsetneq S_{\mathrm{aff}}$ and a unipotent representation $V = V_{(x,\phi)}$ of G, with $x \in G^\vee$ and $\phi \in \widehat{A_x}$. For each irreducible unipotent $\overline{K}_J$-representation χ, we want to calculate the value of $\langle \chi, V^{U_J} \rangle_{K_J}$, where we view the $\overline{K}_J$-representations as K_J-representations trivial on U_J.

In the first part, we state two reduction steps that allow us to reduce the problem of calculating the parahoric restrictions of V to the problem of calculating the restriction of some representation obtained from V over a reflection group with respect to certain reflection subgroups. For example, if V is Iwahori-spherical, we obtain a representation $V^{\mathbf{1}}_{q=1}$ over the extended affine Weyl group $\widetilde{W}$ of G, and the parahoric restriction of V can be determined by studying the representation $V^{\mathbf{1}}_{q=1}|_{W_J}$ where W_J is the subgroup of $\widetilde{W}$ generated by the reflections in J. Springer theory is then the key theoretical tool that allows us to calculate the action of $\widetilde{W}$ on $V^{\mathbf{1}}_{q=1}$. We devote the second part of the chapter to discuss this. However, not every unipotent representation of G is Iwahori-spherical, so in the last part we discuss a more general method to calculate the parahoric restriction of non Iwahori-spherical representations with the use of weight diagrams developed in cite here the other paper of Reeder. These have been computed for all unipotent representations of adjoint exceptional groups in [Ree00, §9].

4.1 Reduction to representations over reflection groups

Consider the fixed unipotent representation $V = V_{(x,\phi)}$ of G. By the classification of unipotent representations there is a unique proper subset I of S_{aff} up to the action of Ω with corresponding standard parahoric $K := K_I$, such that V^{U_K} contains a cuspidal unipotent representation σ of $\overline{K}$. In particular, the vector space

$$V^{\sigma} := \mathrm{Hom}_K(\sigma, V^{U_K}) = \mathrm{Hom}_K(\sigma, V) = \mathrm{Hom}_G(c\text{-}\mathrm{Ind}_K^G \sigma, V) \tag{10}$$

is a non-zero irreducible $\mathcal{H}(G, K, \sigma) = \mathrm{End}_G(c\text{-}\mathrm{Ind}_K^G \sigma)$-module. We say that the block parametrized by the pair (K, σ) is controlled by the Hecke algebra $\mathcal{H}(G, K, \sigma)$.

Similarly, if χ is an irreducible unipotent representation of $\overline{K}_J$, there is a unique proper subset I' of S_{aff} contained in J such that χ is a subrepresentation of $\mathrm{Ind}_{K_{I'}}^{K_J} \sigma'$, where σ' is a cuspidal unipotent representation of $\overline{K}_{I'}$. In particular, the vector space

$$\chi^{\sigma'} := \mathrm{Hom}_{K_{I'}}(\sigma', \chi) = \mathrm{Hom}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma', \chi) \neq 0.$$

The space $\chi^{\sigma'}$ is naturally a $\mathcal{H}(K_J, K_{I'}, \sigma')$-module, where $\mathcal{H}(K_J, K_{I'}, \sigma') \cong \mathrm{End}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma')$ is the subalgebra of functions of $\mathcal{H}(G, K_{I'}, \sigma')$ supported on K_J.

Lemma 4.1 (First reduction)**.** *([Ree00, §4.2]) The multiplicity of the simple K_J-module χ in V^{U_J} is given by*

$$\langle \chi, V^{U_J} \rangle_{K_J} = \langle \chi^{\sigma'}, V^{\sigma'} \rangle_{\mathcal{H}(K_J, K_{I'}, \sigma')}.$$

Moreover, $\langle \chi, V^{U_J} \rangle = 0$ unless I and I' are conjugate under the action of Ω. In particular, if J does not contain any subset Ω-conjugate of I, then $V^{U_J} = 0$.

Proof. Since $U_J \subseteq U_{I'}$, we have that

$$\begin{aligned} V^{\sigma'} = \mathrm{Hom}_{K_{I'}}(\sigma', V) = \mathrm{Hom}_{K_{I'}}(\sigma', V^{U_J}) &\simeq \mathrm{Hom}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma', V^{U_J}) \\ &\simeq \bigoplus_{\eta} \langle \eta, V^{U_J} \rangle_{K_J} \mathrm{Hom}_{K_J}(\mathrm{Ind}_{K_{I'}}^{K_J} \sigma', \eta) \\ &\simeq \bigoplus_{\eta} \langle \eta, V^{U_J} \rangle_{K_J} \eta^{\sigma'}, \end{aligned}$$

and therefore

$$\langle \chi^{\sigma'}, V^{\sigma'} \rangle_{\mathcal{H}(K_J, K_{I'}, \sigma')} = \sum_{\eta} \langle \chi^{\sigma'}, \eta^{\sigma'} \rangle_{\mathcal{H}(K_J, K_{I'}, \sigma')} \langle \eta, V^{U_J} \rangle_{K_J} = \langle \chi, V^{U_J} \rangle_{K_J}.$$

To prove the second statement, we note that $V^{\sigma'} \neq 0$ if and only if (K, σ) is conjugate to $(K_{I'}, \sigma')$. In other words, I and I' are conjugate under the action of Ω. The last assertion follows immediately from the fact that J contains I'. □

Example 4.2. Let us briefly explore the two extreme cases of the above result.

- If V is a Iwahori-spherical representation of G, then $V^{\mathcal{I}} \neq 0$, $I = \emptyset$ and the pair (K, σ) is given by $(\mathcal{I},\mathbf{1})$. In this case, Lemma 4.1 states that

$$\langle \chi, V^{U_J}\rangle_{K_J} = \langle \chi^{\mathcal{I}}, V^{\mathcal{I}}\rangle_{\mathcal{H}(K_J,\mathcal{I},\mathbf{1})}.$$

Thus, V has a non-zero parahoric restriction for any $J \subsetneq S_{\text{aff}}$.

- If V is a supercuspidal unipotent representation of G, then I is a maximal proper subset of S_{aff} and $V^{U_J} = 0$ unless $J = I$.

Now let us consider the second reduction step. For this one, we let $R = \mathbb{C}(v, v^{-1})$, where v is an indeterminate. We then define $\mathcal{H}(G,K,\sigma)_v$ to be the *generic Hecke algebra* defined over R with the same generators and relations as $\mathcal{H}(G,K,\sigma)$, but with q replaced by v^2. The upshot of considering this generic Hecke algebra is that by specializing v we can recover

$$\mathcal{H}(G,K,\sigma)_{\sqrt{q}} = \mathcal{H}(G,K,\sigma), \qquad \mathcal{H}(G,K,\sigma)_1 = \mathbb{C}[\tilde{W}(K,\sigma)],$$

where $\tilde{W}(K,\sigma)$ is an explicit affine reflection group associated to the pair (K,σ) (see [Lus95] for an explicit list, and [Car85, §10] for a detailed construction). When $K = \mathcal{I}$ is the Iwahori subgroup and $\sigma = \mathbf{1}$ is the trivial representation, $\mathcal{H}(G,\mathcal{I},\mathbf{1})_1 = \mathbb{C}[\widetilde{W}]$, where $\widetilde{W}$ is the extended affine Weyl group. Instead, if K is a maximal parahoric subgroup of G, then $\mathcal{H}(G,K,\sigma) \cong \mathbb{C}$ so $\tilde{W}(K,\sigma)$ is trivial.

Fact: For any simple $\mathcal{H}(G,K,\sigma)$-module E *considered in this document*, there is a $\mathcal{H}(G,K,\sigma)_v$-module E_v such that

$$E \simeq E_v \otimes_R \mathbb{C}, \text{ where } f \in R \text{ acts on } \mathbb{C} \text{ by } f(\sqrt{q}).$$

It then follows that we have a $q = 1$ operation that takes modules over Hecke algebras to modules over the corresponding reflection groups, obtained by setting $v = 1$ in all matrix coefficients of the generic module.

Proposition 4.3 (Second reduction). *(*[Ree00, §6.1]*) Let $J \subsetneq S_{\text{aff}}$ and let $(K, U_K, \overline{K})$ be a parahoric subgroup contained in $(K_J, U_J, \overline{K}_J)$ with cuspidal unipotent $\overline{K}$-representation σ. Then the diagram*

$$\begin{array}{ccc}
\mathcal{H}(G,K,\sigma)\text{-}mod & \xrightarrow{q=1} & \tilde{W}(K,\sigma)\text{-}mod \\
{\scriptstyle \mathrm{Res}_{\mathcal{H}(K_J,K,\sigma)}}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{Res}_{\tilde{W}(K,\sigma)_J}} \\
\mathcal{H}(K_J,K,\sigma)\text{-}mod & \xrightarrow{q=1} & \tilde{W}(K,\sigma)_J\text{-}mod
\end{array}$$

is commutative, and the bottom arrow is an isometry with respect to the usual inner product of character. That is, for any irreducible K_J-module χ, we have that

$$\langle \chi^\sigma, V^\sigma\rangle_{\mathcal{H}(K_J,K,\sigma)} = \langle \chi^\sigma_{q=1}, V^\sigma_{q=1}\rangle_{\tilde{W}(K,\sigma)_J}. \tag{11}$$

Combining both reduction steps, we obtain the identity

$$\langle \chi, V^{U_J} \rangle_{K_J} = \langle \chi^\sigma, V^\sigma \rangle_{\mathcal{H}(K_J, K, \sigma)} = \langle \chi^\sigma_{q=1}, V^\sigma_{q=1} \rangle_{\tilde{W}(K,\sigma)_J} \tag{12}$$

Thus, to compute the parahoric restriction V^{U_J}, it suffices to study the $\tilde{W}(K,\sigma)_J$-module $V^\sigma_{q=1}$. By calculating its decomposition into $\tilde{W}(K,\sigma)$-irreducible representation, one can directly obtain the decomposition of V^{U_J} into irreducible $\overline{K}_J$-representations.

At this point, we restrict our attention to the parahoric restriction of Iwahori-spherical representations, for Springer theory provides a way to calculate the $K(I, \mathbf{1})_J$-module $V^{\mathbf{1}}_{q=1}$ explicitly, differing significantly from the non Iwahori-spherical case. As stated above, for Iwahori-spherical representations, the reflection subgroup $\tilde{W}(\mathcal{I}, \mathbf{1}) \cong \widetilde{W}$ is simply the extended affine Weyl group of G, and $\tilde{W}(\mathcal{I}, \mathbf{1}) = \widetilde{W}_J$ is the subgroup of $\widetilde{W}$ generated by the simple reflections of the roots in J.

4.2 The Iwahori-spherical case and the Springer correspondence

In this subsection, let us fix an Iwahori-spherical representation $V = V_{(x,\phi)}$ of G and some proper subset $J \subsetneq S_{\text{aff}}$. The two reduction steps stated above imply that for any principal series representation χ of the reductive quotient $\overline{K}_J$,

$$\langle \chi, V^{U_J} \rangle_{\overline{K}_J} = \langle \chi^{\mathbf{1}}_{q=1}, V^{\mathbf{1}}_{q=1} \rangle_{\widetilde{W}_J}. \tag{13}$$

Springer theory provides the tools to explicitly describe the structure of the $\widetilde{W}_J$-module $V^{\mathbf{1}}_{q=1}$ in terms of the geometry of the dual group $G^\vee$ and the data (x, ϕ). To extract this information, consider the Jordan decomposition $x = us$, with commuting $u \in G^\vee$ unipotent and $s \in G^\vee$ semisimple. Let $\mathcal{B}$ and $\mathcal{B}_s$ be the flag varieties of the complex reductive groups $G^\vee$ and $G^\vee_s := Z_{G^\vee}(s)$, respectively. These are algebraic varieties parametrizing the set of Borel subgroups in $G^\vee$ and $G^\vee_s$, respectively, and admit a rational action by conjugation. We then consider the stabilizers $\mathcal{B}^x$ and $\mathcal{B}^u_s$, called partial flag varieties or Springer fibers. These are algebraic subvarieties parametrizing the set of Borel subgroups of $G^\vee$ containing x and the set of Borel subgroups of $G^\vee_s$ containing u.

The varieties $\mathcal{B}_x$ and $\mathcal{B}^u_s$ admit a natural action of A_x induced by the action of $Z_{G^\vee}(x)$ by conjugation. While these varieties do not admit a natural action of the Weyl groups W and W_s of $G^\vee$ and $G^\vee_s$, respectively, Springer's celebrated result states that the singular cohomology spaces $H^*(\mathcal{B}_x)$ and $H^*(\mathcal{B}^u_s)$ do admit a natural action of W and W_s, and that both actions commute with the A_x-action inherited from the action on the partial flag varieties. These last two statements are at the heart of Springer theory, which we shall use continuously from now on. The mathematics behind these results involve sophisticated geometric machinery, so for the most part we will state the results we need without proof. The Springer correspondence lies at the top of Springer theory, which we now state for convenience.

Theorem 4.4. *Let $G^\vee$ be a complex algebraic reductive group with Weyl group W. For each pair (u, ϕ),* *where $u \in G^\vee$ is a unipotent element and ϕ is an irreducible character of A_u, the ϕ-isotropic subspace of the top cohomology group $H^{\mathrm{top}}(\mathcal{B}^u)^\phi$ is either zero or an irreducible W-representation. Moreover, each irreducible character of W arises this way for exactly one pair (u, ϕ) up to $G^\vee$ conjugation. In other words, there is an injection*

$$\mathrm{Irr}(W) \hookrightarrow \{\textit{pairs } (u, \phi)\}/G^\vee.$$

Moreover, for any unipotent $u \in G^\vee$, the pair $(u, \mathbf{1})$ lies in the image of the map above.

This result is of great importance for it allows us to extract information from the top cohomology group directly. When $G^\vee$ is a classical group, one can give precise description of the injection in terms of combinatorial data. When $G^\vee$ is of exceptional type, one only has a finite amount of information and the injection above can be found in the literature (for example, in [Car85, §13.3]). Another important fact that greatly simplifies the calculations is the following:

Theorem 4.5 ([KL87])**.** *The odd-degree cohomology spaces of $\mathcal{B}^x$ vanish, so $H^*(\mathcal{B}^x) = \sum_{i=1}^{\dim \mathcal{B}^x} H^{2i}(\mathcal{B}^x)$.*

Example 4.6. If $G^\vee = \mathrm{GL}_n(\mathbb{C})$, by the Jordan decomposition theorem, unipotent conjugacy classes are parametrized by partitions of n and $A_u = \{1\}$ for any unipotent element u. Moreover, $W \cong S_n$, and its irreducible representations are also labelled by partitions of n. If λ is a partition of n, then the Springer correspondence maps V_λ to $(u_\lambda, \mathbf{1})$, where V_λ is the Specht module of S_n and u_λ is a unipotent element of $\mathrm{GL}_n(\mathbb{C})$ corresponding to λ. In particular, the Springer correspondence for $\mathrm{GL}_n(\mathbb{C})$ is a bijection. On the other hand, if $G^\vee = \mathrm{SL}_n(\mathbb{C})$, conjugacy classes of unipotent elements are still parametrized by partitions of n. This time, however, A_u may be non-trivial for some unipotent classes. The Springer correspondence is therefore not a bijection, whose the image consists of all pairs $\{(u, \mathbf{1}), u \in \mathrm{SL}_n(\mathbb{C}) \text{ unipotent}\}$.

When $G^\vee$ has type A_n, one can describe the entire cohomology complex $H^*(\mathcal{B}^u)^{\mathbf{1}}$ explicitly. Suppose that $u = u_\lambda$ for a partition $\lambda = (\lambda_1, \ldots, \lambda_r)$ of n. If $S_\lambda = S_{\lambda_1} \times \cdots \times S_{\lambda_r} \leq S_n$, then Lusztig and Spaltenstein showed

$$H^*(\mathcal{B}^{u_\lambda}) \cong \mathrm{Ind}_{S_\lambda}^{S_n} \mathbf{1}$$

as graded S_n-modules.

Example 4.7. The complex reductive group $G^\vee = G_2(\mathbb{C})$ has Weyl group $W \cong D_6$ and 5 unipotent classes labelled and partially ordered by

$$1 < A_1 < \tilde{A}_1 < G_2(a_1) < G_2.$$

The component groups are all trivial except for the subregular unipotent class, whose component group is isomorphic to S_3, with irreducible representations $\mathbf{1}, \varepsilon$ and r (the unique 2-dimensional). The characters of W are labelled in the literature by the symbols $\phi_{1,0}$ (trivial), $\phi_{1,6}$ (sign), $\phi'_{1,3}, \phi''_{1,3}$ (the other two characters), $\phi_{2,1}$ (faithful) and $\phi_{2,2}$ (lifted from S_3). Then the Springer correspondence gives the pairing

$$\phi_{1,0} \leftrightarrow (G_2, \mathbf{1}), \quad \phi_{2,1} \leftrightarrow (G_2(a_1), \mathbf{1}), \quad \phi'_{1,3} \leftrightarrow (G_2(a_1), r), \quad \phi_{2,2} \leftrightarrow (\tilde{A}_1, \mathbf{1}), \quad \phi''_{1,3} \leftrightarrow (A_1, \mathbf{1}), \quad \phi_{1,6} \leftrightarrow (1, \mathbf{1}),$$

while the pair $(G_2(a_1), \varepsilon)$ is not in the image of the correspondence. This is explained by the fact that $(G_2(a_1), \varepsilon)$ is a *cuspidal local system* on $G_2(\mathbb{C})$.

Having explained the Springer correspondence, we can go back to our initial discussion. We now know that $H^*(\mathcal{B}^x)$ has the structure of a $A_x \times W$-module. We can extend this to a $A_x \times \widetilde{W}$-action as follows. The extended Weyl group can be decomposed as a semidirect product $\widetilde{W} = W \ltimes X$, where X is the character lattice of $T^\vee$. Then there is a natural evaluation pairing $s : X \to \mathbb{C}^\times$ given by $\mu \mapsto \langle s, \mu \rangle$. The character lattice X then acts on $H^*(\mathcal{B}^x)$ by scalars $\langle s, \cdot \rangle$, and this extends the action as desired. Analogously, the $A_x \times W_s$-action of $H^*(\mathcal{B}^u_s)$ can be extended to a $A_x \times \widetilde{W}_s$-action. The main result we need is due to Lusztig.

Theorem 4.8. *Let $V_{x,\phi}$ be an Iwahori-spherical irreducible representation of G. After letting $q \to 1$ and then taking semisimplification, the simple $\mathcal{H}(G, I, \mathbf{1})$-module $V^{\mathbf{1}}_{x,\phi}$ becomes the $\widetilde{W}$-module $\varepsilon \otimes H^*(\mathcal{B}^x)^\phi$.*

Remark 4.9. An important remark is due at this point. The above theorem relates the structure of $V^{\mathbf{1}}_{(x,\phi),q=1}$ as a $\widetilde{W}$-module with the structure of $H^*(\mathcal{B}^x)^\phi$ as a $\widetilde{W}$-module too. However, the former object is naturally p-adic, while the latter is complex analytic. The Weyl groups of G and $G^\vee$ are canonically isomorphic, but not equal if the group is not simply laced, since the isomorphism swaps short with long roots. It is a standard abuse of notation to denote both Weyl groups with the same symbol, but one needs to be careful during explicit calculations inside which Weyl group one is working. We are ultimately interested in the p-adic side, but most computations involving cohomology groups and the Springer correspondence take place in the complex side. In particular, the reflection s_0 inside W will be a **short** reflection, corresponding to the highest short root. At the very end, we will then trace the representations back to the p-adic side by swapping long and short roots.

To conclude this section, we note that the element $x \in G^\vee$ parametrizing $V_{x,\phi}$ is not necessarily unipotent, so the Springer correspondence does not directly apply to $H^*(\mathcal{B}^x)^\phi$. However, the Jordan decomposition $x = us$ allows us to relate $H^*(\mathcal{B}^x)^\phi$ to $H^*(\mathcal{B}^u_s)^\phi$, which is a $\widetilde{W}_s$-module that can be described by Springer theory. The result is due to Kato.

Proposition 4.10. [Kat83] *The natural restriction map $H^*(\mathcal{B}^x)^\phi \to H^*(\mathcal{B}^u_s)^\phi$ is $\widetilde{W}_s$-equivariant, and induces an isomorphism of $\widetilde{W}$-modules*

$$H^*(\mathcal{B}^x)^\phi \cong \mathrm{Ind}_{\widetilde{W}_s}^{\widetilde{W}} H^*(\mathcal{B}^u_s)^\phi.$$

By (13), Theorem 4.8 and Proposition 4.10, the restriction of $V_{x,\phi}$ can be determined by computing the restriction

$$\left(\mathrm{Ind}_{\widetilde{W}_s}^{\widetilde{W}} H^*(\mathcal{B}_s^u)^\phi\right)|_{\widetilde{W}_J},$$

where $\widetilde{W}_J$ is the subgroup of $\widetilde{W}$ generated by all simple reflections of the roots in J. Working inside $\widetilde{W}$ is delicate due to the action of the semisimple element s on the character lattice, so following the development in [Ree00, §8], we work inside $W = \widetilde{W}_{J_0}$. The downside is the fact that, even though W contains an isomorphic copy of $\widetilde{W}_J$, one needs to carefully track this isomorphism since it can twist the original representation by some character.

More concretely, consider the image W_J of the group $\widetilde{W}_J$ under the natural projection map $\widetilde{W} \twoheadrightarrow W$. This restricts to an isomorphism $\widetilde{W}_J \xrightarrow{\sim} W_J$ with inverse map $\psi_J : W_J \to \widetilde{W}_J$, satisfying $\Psi_J(s_\alpha) = s_\alpha$ if $\alpha \in J, \alpha \neq -\alpha_0$ and $\psi_J(s_0) = \tilde{s}_0 = t_{\alpha_0} s_0$ if $\tilde{s}_0 \in J$. Let ψ_J^* be the pullback of representations of $\widetilde{W}_J$ to W_J.

Given a semisimple element $t \in G^\vee$, let $W_{J,t} = W_t \cap W_J$ and define the character

$$\chi_t^J := \chi_t \circ \psi_J : W_{J,t} \longrightarrow \mathbb{C}^\times. \tag{14}$$

Then by Mackey theory we obtain the isomorphism

$$\psi_J^*\left[\left(\mathrm{Ind}_{\widetilde{W}_s}^{\widetilde{W}} H^*(\mathcal{B}_s^u)^\phi\right)|_{\widetilde{W}_J}\right] \cong \bigoplus_{w \in W_s \backslash W / W_J} \mathrm{Ind}_{W_{J,s^w}}^{W_J} \chi_{s^w}^J \otimes \left[H^*(\mathcal{B}_s^u)^\phi\right]^w. \tag{15}$$

Springer theory tells us the structure of the cohomology groups, so it remains to understand the groups $W_{J,t}$ and the characters χ_t^J. The groups $W_{J,t}$ have been studied in the literature, and they can be decomposed into

$$W_{J,t} = W_{J,t}^0 \rtimes R_{J,t},$$

where $W_{J,t}^0$ is a "large" reflection subgroup of W generated by the reflections in J that fix t, and $R_{J,t}$ is a "small" group that normalizes $W_{J,t}^0$. The character χ_t^J is trivial on $W_{J,t}^0$ and is determined by its restriction to $R_{J,t}$. As a result, the characters χ_t^J are trivial in any cases. For example, if $W_{J,t} = W_{J,t}^0$ is a reflection subgroup of W, then χ_t^J is trivial. In addition, one can associate to J a subgroup Z_J of the center of the simply connected cover of a pseudo-Levi subgroup $G_J^\vee$ generated by the torus and the roots J, and if the order of t is relatively prime to the order of Z_J, then χ_t^J is trivial as well. We refer to [Ree00, §8] for a detailed discussion on the structure of these characters, where explicit formulas to compute them are provided. All nontrivial characters in this paper have been calculated using that formula. If all characters $\chi_{s^w}^J$ appearing in (15) are trivial, then computations are greatly simplified, as the next result shows.

Lemma 4.11. *With the same notation as in* (15)*, if the characters $\chi_{s^w}^J$ are trivial for all $w \in W_s \backslash W / W_J$, then*

$$\psi_J^*\left[\left(\mathrm{Ind}_{\widetilde{W}_s}^{\widetilde{W}} H^*(\mathcal{B}_s^u)^\phi\right)|_{\widetilde{W}_J}\right] \cong \left(\mathrm{Ind}_{W_s}^{W} H^*(\mathcal{B}_s^u)^\phi\right)|_{W_J}$$

Proof. This is an immediate consequence of Mackey's formula applied to the right hand side. □

4.3 Non-Iwahori-spherical case and weight diagrams

For this section, let us fix a non-Iwahori-spherical representation $V_{x,\phi}$ of G with associated parahoric $K := K_I$ and cuspidal unipotent representation σ as in (10). Then V^σ is a simple module over the Hecke algebra $\mathcal{H}(G, K, \sigma) = \mathrm{End}_G(c\text{-}\mathrm{Ind}_K^G \sigma)$. The structure of the Hecke algebras have been widely studied in the literature (see [Lus95, §1] for a detailed discussion). In particular, one can associate a canonical extended affine Weyl group $\tilde{W}(K, \sigma)$ parametrizing a basis $\{T_w, w \in \tilde{W}(K, \sigma)\}$ to the Hecke algebra $\mathcal{H}(G, K, \sigma)$ satisfying explicit relations. The structure of a modules over $\mathcal{H}(G, K, \sigma)$ can then be largely understood by studying the action of the basis elements T_s corresponding to simple reflections $s \in \tilde{W}(K, \sigma)$. Importantly, these elements satisfy the quadratic relations

$$(T_s + 1)(T_s - q^{c(s)}) = 0,$$

where $c(s)$ is an explicit positive integer depending on the simple root corresponding to s.

On the other hand, the extended affine Weyl group $\tilde{W}(K, \sigma)$ can be decomposed as a semidirect product $W(K, \sigma) \ltimes X(K, \sigma)$, where $W(K, \sigma)$ can be realized as the Weyl group of an associated complex reductive group $\mathbf{G}$ and $X(K, \sigma)$ as the character lattice of a maximal torus $\mathbf{T}$ of $\mathbf{G}$. The subalgebra spanned by $\{T_w, w \in W(K, \sigma)\}$ is naturally isomorphic to the subalgebra $\mathcal{H}(K_{J_0}, K, \sigma)$ of functions supported on a hyperspecial maximal compact subgroup K_{J_0} of G. We then have an explicit isomorphism (cite Solleveld)

$$\mathcal{H}(G, K, \sigma) \cong \mathcal{H}(K_{J_0}, K, \sigma) \mathbin{\tilde{\otimes}} \mathcal{O}[\mathbf{T}], \tag{16}$$

where $\tilde{\otimes}$ is a twisted tensor product of $\mathbb{C}$-vector space with a twisted multiplication given by

$$(T_s - q^{c(s)})f - f^s(T_s - q^{c(s)}) = (f^s - f)\zeta_s \in \mathcal{O}[\mathbf{T}]$$

where $\zeta_s \in (f^s - f)\mathcal{O}[\mathbf{T}]$ is a rational function on $\mathbf{T}$.

Example 4.12. The only non-trivial example we encounter in this paper is the case of $G = F_4(F)$, K a parahoric subgroup of type B_2 and σ the unique cuspidal unipotent representation of $B_2(\mathbb{F}_q)$. In this case, $\tilde{W}(K, \sigma)$ is the affine Weyl group of type C_2 and the Hecke algebra $\mathcal{H}(G, K, \sigma)$ is generated by the elements $T_{s_0}, T_{s_1}, T_{s_2}$ satisfying the quadratic relations

$$(T_{s_0} + 1)(T_{s_0} - q) = 0, \quad (T_{s_i} + 1)(T_{s_i} - q^3) = 0, \quad \text{for } i = 1, 2.$$

Moreover, $\mathbf{G} = \mathrm{SO}_5(\mathbb{C})$ and $\mathbf{T} = (\mathbb{C}^\times)^2$, with simple roots

$$e_{\alpha_1}(t_1,t_2) = t_1/t_2, \quad e_{\alpha_2}(t_1,t_2) = t_2,$$
$$\zeta_{s_1} = \frac{q^3 - e_{\alpha_1}}{1 - e_{\alpha_1}}, \quad \zeta_{s_2} = \frac{(q^2 - e_{\alpha_2})(q + e_{\alpha_2})}{1 - e_{2\alpha_2}}.$$

The theory of weight diagrams developed by Reeder in (Introduce his reference here) studies first the restriction of the module V^σ to the large subalgebra $\mathcal{O}[\mathbf{T}]$ and then the action of the basis elements T_s for $s \in W(K,\sigma)$ a simple reflection on the resulting decomposition. Since $\mathcal{O}[\mathbf{T}]$ is a commutative algebra, we can also decompose

$$V^\sigma|_{\mathcal{O}[\mathbf{T}]} = \bigoplus_{\tau \in \mathbf{T}} V^\sigma(\tau),$$

where $V^\sigma(\tau)$ consists of those vectors in V^σ killed by some power of the maximal ideal of $\mathcal{O}[\mathbf{T}]$ corresponding to τ. From this information, one can then determine the action of the basis elements $W(K,\sigma)$ on the space $V^\sigma_{q=1}$, which immediately gives the parahoric restriction of V with respect to the maximal hyperspecial parahoric subgroup K_{J_0}. See [Ree00, §9.5] for the explicit formulas. To obtain the parahoric restriction with respect to other parahoric subgroups, one needs to understand the action of the remaining simple reflection $s_0 \in \tilde{W}(K,\sigma)$. This can be done by tracing back through the isomorphism $\mathcal{H}(G,K,\sigma) \cong \mathcal{H}(K_{J_0},K,\sigma) \tilde{\otimes} \mathcal{O}[\mathbf{T}]$ and using the explicit formula for the twisted multiplication. This can be a lengthy calculation, so we will omit them and only provide the final results without proof.

4.4 Parahoric restriction for type G_2

We end this section with the explicit calculation of the parahoric restrictions for some representations $\pi(x,\phi)$ of the p-adic group $G = G_2(F)$, highlighting the methods described above with a concrete example. The group $G_2(F)$ has simple roots $\{-\beta_0, \beta_1, \beta_2\}$ with Dynkin diagram

$$0 - 1 \Longrightarrow 2.$$

Thus $G_2(F)$ has 3 classes of maximal parahoric subgroups K_0, K_1, K_2 corresponding to the subsets of affine simple roots $J_0 = \{\beta_1, \beta_2\}$, $J_1 = \{-\beta_0, \beta_2\}$ and $J_2 = \{-\beta_0, \beta_1\}$ and with reductive types G_2, $A_1 \times \tilde{A}_1$ and A_2, respectively. We now compute these parahoric restrictions for all representations $\pi(us,\phi)$ of $G = G_2(F)$, where u is a subregular unipotent element of $G_2(\mathbb{C})$ labelled by $G_2(a_1)$. This is an interesting case since u is a distinguished unipotent element and $\Gamma_u = A_u \cong S_3$. There are 8 such representations, given by

$$\pi(G_2(a_1),1,\phi), \phi \in \{\mathbf{1},\varepsilon,r\}, \quad \pi(G_2(a_1),g_2,\phi), \phi \in \{\mathbf{1},\varepsilon\}, \quad \pi(G_2(a_1),g_3,\phi), \phi \in \{\mathbf{1},\theta,\theta^2\}.$$

Lusztig's arithmetic-geometric correspondence directly implies that 4 of them are unipotent supercuspidal representations of $G_2(F)$ given by

$$\pi(G_2(a_1), 1, \epsilon) = c\text{-Ind}_{K_0}^{G} G_2[1], \quad \pi(G_2(a_1), g_2, \epsilon) = c\text{-Ind}_{K_0}^{G} G_2[-1] \quad \text{and}$$
$$\pi(G_2(a_1), g_3, \theta^k) = c\text{-Ind}_{K_0}^{G} G_2[\theta^k], k = 1, 2,$$

while the remaining 4 are Iwahori-spherical. For the supercuspidal representations, their restrictions can be easily computed. By Frobenius reciprocity, their parahoric restrictions to K_0 is irreducible. The restrictions to K_1 and K_2 are both 0, since all representations of $A_1 \times \tilde{A}_1$ and A_2 are principal series.

Thus, we can focus our attention to Iwahori-spherical representations. The computations are quite lenghty, so we give a complete sketch while omitting some unenlightening steps. We recall that we are working inside the *complex* Weyl group W and consequently s_0 is seen as a short reflection along the highest short root, and we denote by s^0 the reflection along the highest root. We first note that u is a subregular unipotent element in $G^\vee = G_2(\mathbb{C})$, while it is a regular unipotent element in $G^\vee_{g_2} = \mathrm{GL}_2(\mathbb{C})$ and $G^\vee_{g_3} = \mathrm{SL}_3(\mathbb{C})$. Therefore, $\dim \mathcal{B}^u = 1$, while $\dim \mathcal{B}^u_{g_2} = \dim \mathcal{B}^u_{g_3} = 0$. By Springer theory, we obtain that

$$H^*(\mathcal{B}^u)^{\mathbf{1}} = H^0(\mathcal{B}^u)^{\mathbf{1}} + H^2(\mathcal{B}^u)^{\mathbf{1}} = \phi_{1,0} + \phi_{2,1}, \quad \text{and} \quad H^*(\mathcal{B}^u)^r = H^0(\mathcal{B}^u)^r + H^2(\mathcal{B}^u)^r = \phi'_{1,3},$$

as W-modules, while $H^0(\mathcal{B}^u_{g_2})$ and $H^0(\mathcal{B}^u_{g_3})$ afford the trivial representations of W_{g_2} and W_{g_3} respectively, and

$$\mathrm{Ind}_{W_{g_2}}^{W} \mathbf{1} = \phi_{1,0} + \phi''_{1,3} \quad \text{and} \quad \mathrm{Ind}_{W_{g_3}}^{W} \mathbf{1} = \phi_{1,0} + \phi_{2,2}.$$

To obtain the parahoric restrictions to K_0, we note that all characters $\chi_t^{J_0}$ are trivial so it remains to twist by $\varepsilon = \phi_{1,6}$ (swapping $\phi_{1,0} \leftrightarrow \phi_{1,6}$ and $\phi'_{1,3} \leftrightarrow \phi''_{1,3}$) and then by the outer automorphism of W exchanging short and long roots (swapping $\phi'_{1,3} \leftrightarrow \phi''_{1,3}$ only).

To compute the parahoric restrictions with respect to K_1 and K_2 in Table 1 we compute the remaining characters χ_t^J. Using their properties above, it is not hard to see that all characters we are considering are trivial except for the pairs $(s, J) = (g_2, J_1)$ and $(s, J) = (g_3, J_2)$. When the characters are trivial, Lemma 4.11 and Alvis' tables immediately yield the results. The last two cases must be dealt with separately.

- **Restriction of** $\pi(G_2(a_1), g_3, 1)$ **to** K_{J_2}**.** In this case, $W_{J_2} = \langle s_0, s_1 \rangle$ (where s_0 is the reflection along the highest short root) and $W_{g_3} = \langle s^0, s_2 \rangle$ (where s^0 is the reflection along the highest root). Then $W_{g_3} \backslash W / W_{J_2} = \{1\}$, $W_{J_2, g_3} \cong C_3$ and $\chi_{g_3}^{J_2}$ is a primitive character. Thus,

$$\mathrm{Ind}_{W_{J_2, g_3}}^{W_{J_2}} \chi_{g_3}^{J_2} = r.$$

- **Restriction of** $\pi(G_2(a_1), g_2, 1)$ **to** K_{J_1}**.** In this case, $W_{J_1} = \langle s_0, s_2 \rangle$ and $W_{g_2} = \langle s^0, s_1 \rangle$. Then

$W_{g_2}\backslash W/W_{J_1} = \{1, w\}$ has two elements, and $W_{J_1,g_2} \cong C_2$ with $\chi^{J_1}_{g_2}$ non-trivial and $W_{J_1,g_2^w} = W_{J_1}$ with $\chi^{J_1}_{g_2^w}$ the trivial character. Thus,

$$\mathrm{Ind}^{W_{J_1}}_{W_{J_1,g_2}} \chi^{J_1}_{g_2} + \mathrm{Ind}^{W_{J_1}}_{W_{J_1,g_2^w}} \chi^{J_1}_{g_2^w} = \varepsilon \boxtimes \mathbf{1} + \mathbf{1} \boxtimes \varepsilon + \mathbf{1} \boxtimes \mathbf{1}.$$

Twisting by the sign character yields the desired result. The proof of Conjecture 1.1 is now just a matter of putting all the pieces together. The parahoric restrictions of the virtual characters $\Pi(u, s, h)$ can be computed by linearity from the parahoric restrictions of the representations $\pi(u, s, \phi)$, which are given in Table 1.

	$K_0 \longrightarrow G_2$	$K_1 \longrightarrow A_1 \times \tilde{A}_1$	$K_2 \longrightarrow A_2$
$\Pi(G_2, 1, 1)$	$\phi_{1,6}$	$\epsilon \boxtimes \epsilon$	ϵ
$\Pi(G_2(a_1), 1, 1)$	$\phi_{2,1} + G_2[1] + 2\phi'_{1,3} + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + 3\epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$3\epsilon + r$
$\Pi(G_2(a_1), 1, g_2)$	$\phi_{2,1} - G_2[1] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$\epsilon + r$
$\Pi(G_2(a_1), g_2, 1)$	$\phi_{2,1} + \phi_{2,2} + G_2[-1] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$\epsilon + r$
$\Pi(G_2(a_1), 1, g_3)$	$\phi_{2,1} + G_2[1] - \phi'_{1,3} + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r
$\Pi(G_2(a_1), g_3, 1)$	$\phi''_{1,3} + G_2[\theta] + G_2[\theta^2] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r
$\Pi(G_2(a_1), g_2, g_2)$	$\phi_{2,1} + \phi_{2,2} - G_2[-1] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \mathbf{1} + \epsilon \boxtimes \epsilon$	$\epsilon + r$
$\Pi(G_2(a_1), g_3, g_3)$	$\phi''_{1,3} + \theta G_2[\theta] + \theta^2 G_2[\theta^2] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r
$\Pi(G_2(a_1), g_3, g_3^{-1})$	$\phi''_{1,3} + \theta^2 G_2[\theta] + \theta G_2[\theta^2] + \phi_{1,6}$	$\mathbf{1} \boxtimes \epsilon + \epsilon \boxtimes \epsilon$	r

Table 1: Restrictions of G_2-representations $\pi(G_2(a_1), s, \phi)$

Finally, Lusztig's nonabelian Fourier transform acts trivially on every representation, except for the $G_2(\mathbb{F}_q)$ family

$$\{\phi_{2,1}, \phi'_{1,3}, G_2[1], \phi_{2,2}, G_2[-1], \phi''_{1,3}, G_2[\theta], G_2[\theta^2]\},$$

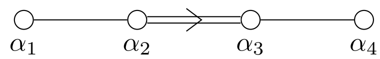
where the action is given by the 8×8 matrix It is now a routine calculation to see that the square in

$$\begin{bmatrix}
\frac{1}{6} & \frac{1}{3} & \frac{1}{6} & \frac{1}{2} & \frac{1}{2} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\
\frac{1}{3} & \frac{2}{3} & \frac{1}{3} & 0 & 0 & -\frac{1}{3} & -\frac{1}{3} & -\frac{1}{3} \\
\frac{1}{6} & \frac{1}{3} & \frac{1}{6} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\
\frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 \\
\frac{1}{2} & 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\
\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} & 0 & 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\
\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} & 0 & 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\
\frac{1}{3} & -\frac{1}{3} & \frac{1}{3} & 0 & 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3}
\end{bmatrix}$$

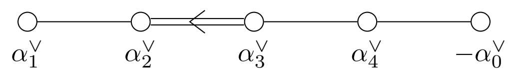
Conjecture 1.1 commutes.

5 The p-adic group of type F_4

In this chapter, we let $G = F_4(F)$ be the simple p-adic group of type F_4 of simply connected type. Since the fundamental group Ω is trivial, F_4 is also adjoint. This is an important fact that simple groups of type F_4 share with simple groups of type G_2 that simplifies many results and calculations. Its dual group is $G^\vee = F_4(\mathbb{C})$, and we let $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$ be the simple roots of $G^\vee$, with Dynkin diagram



We denote by α_0 and α^0 the highest short root and highest root of the root system, respectively, so the simple affine roots of G are $S_{\mathrm{aff}} = \{-\breve{\alpha}_0, \breve{\alpha}_4, \breve{\alpha}_3, \breve{\alpha}_2, \breve{\alpha}_1\}$ with affine Dynkin diagram



Since G is simply connected, it has no pure inner twists other than itself. Moreover, the set of maximal open compact subgroups of G agrees with the set of maximal parahoric subgroups of G, and their conjugacy classes can be parametrized by proper maximal subsets $J \subsetneq S_{\mathrm{aff}}$. It is then easy to see that there are five maximal open compact subgroups $\{K_0, K_4, K_3, K_2, K_1\}$ of G up to conjugacy, where K_i corresponds to $J_i := S_{\mathrm{aff}} \setminus \{\alpha_i^\vee\}$ for $0 \leq i \leq 4$. The reductive quotients $\overline{K}_0, \overline{K}_4, \overline{K}_3, \overline{K}_2, \overline{K}_1$ are finite reductive groups of type $F_4, A_1 \times C_3, A_2 \times \tilde{A}_2, A_3 \times \tilde{A}_1, B_4$, respectively. In particular,

$$\mathcal{C}(G)_{\mathrm{cpt,un}} = \bigoplus_{i=0}^{4} R_{\mathrm{un}}(\overline{K}_i)$$

and $\mathrm{FT}^{\mathrm{par}} = (\mathrm{FT}^{K_i})_i$ is Lusztig's Fourier transform on $R_{\mathrm{un}}(\overline{K}_i)$ on each coordinate.

The aim of this section is to prove that Conjecture 1.1 holds for G. In other words, we prove that

Theorem 5.1. *Let G be a simple p-adic group of type F_4. Then the diagram*

$$\begin{tikzcd}
\mathcal{R}^p_{\mathrm{un,ell}}(G) \arrow[r, "\mathrm{FT}^\vee_{\mathrm{ell}}"] \arrow[d, "\mathrm{res}^{\mathrm{par}}_{\mathrm{un}}"] & \mathcal{R}^p_{\mathrm{un,ell}}(G) \arrow[d, "\mathrm{res}^{\mathrm{par}}_{\mathrm{un}}"] \\
\bigoplus_{i=0}^5 R_{\mathrm{un}}(\overline{K}_i) \arrow[r, "(\mathrm{FT}^{K_i})_i"] & \bigoplus_{i=0}^5 R_{\mathrm{un}}(\overline{K}_i),
\end{tikzcd}$$

commutes.

We remark that the above result was already proven by Ciubotaru, in an unpublished paper [Ciu20] for the reduction to the hyperspecial parahoric subgroup K_0. We complete the proof for the remaining 4 parahoric subgroups.

Firstly, we need to compute the basis $\{\Pi(u,s,h) \mid u \in G^\vee, (s,h) \in \Gamma_u \backslash \mathcal{Y}(\Gamma_u)_{\mathrm{ell}}\}$ of the space $\mathcal{R}^p_{\mathrm{un,ell}}(G)$. Here u is a unipotent element of $G^\vee = F_4(\mathbb{C})$, the complex simple reductive group of type F_4, and $\mathcal{Y}(\Gamma_u)_{\mathrm{ell}}$ is the set of elliptic pairs of Γ_u (see Section should explain again the nonabelian Fourier transform for the Definitions). The classification of all elliptic pairs of $F_4(\mathbb{C})$ has been previously computed, and the list is given in Table 2. For each unipotent $u \in F_4(\mathbb{C})$, we also give the unipotent $F_4(\mathbb{F}_q)$-family $\mathcal{F}_u$ whose special element is the W-representation afforded by $\varepsilon \otimes H^{\mathrm{top}}(\mathcal{B}^u)^{\mathbf{1}}$ under the Springer correspondence. Then $\Gamma_{\mathcal{F}_u}$ is the finite group canonically associated to the family $\mathcal{F}_u$.

Unipotent	Γ_u^0	A_u	$\lvert\Gamma_u\backslash(\Gamma_u)_{\mathrm{ell}}\rvert$	$\mathcal{F}_u$	$\Gamma_{\mathcal{F}_u}$
F_4	1	1	1	$\phi_{1,24}$	1
$F_4(a_1)$	1	S_2	4	$\phi_{4,13}$	C_2
$F_4(a_2)$	1	S_2	4	$\phi_{9,10}$	1
B_3	PGL(2)	1	1	$\phi''_{8,9}$	1
$F_4(a_3)$	1	S_4	21	$\phi_{12,4}$	S_4

Table 2: Elliptic pairs for F_4

In particular, $\mathcal{R}^p_{\mathrm{un,ell}}(G)$ is a 31-dimensional vector space, where 21 of these dimensions come from the distinguished unipotent class $F_4(a_3)$. To prove Theorem 5.1 we need an explicit computation of the restriction map

$$\mathrm{res}^{\mathrm{par}}_{\mathrm{un}} : \mathcal{R}^p_{\mathrm{un,ell}}(G) \longrightarrow \bigoplus_{i=0}^{5} R_{\mathrm{un}}(\overline{K}_i).$$

This takes a large amount of work (we have to compute 155 parahoric restrictions, with varying amounts of effort), and the results are tabulated at the end of the section. We remark that most of this work was completed by Reeder in [Ree00]. Nevertheless, his tables contain a few mistakes (systematically arising from the two representations $\pi(F_4(a_3),1,31)$ and $\pi(F_4(a_3),1,211)$) and do not include the representations whose unipotent label u has type B_3 (for this is not a distinguished unipotent element of $F_4(\mathbb{C})$). Our tables fix these mistakes and add the new restrictions. To display this information for all unipotent classes simultaneously, we use the notation for the Langlands parameters as in [Ree00], which we now explain for convenience.

Notation: Given a representation $\pi(x,\phi)$, let $x = us$ be the Jordan decomposition. Then $u \in Z_{G^\vee}(s)^0$ is a unipotent element of the connected pseudo-Levi subgroup $Z_{G^\vee}(s)^0$, and ϕ is an irreducible representation of the component group A_{su}. The representation $\pi(x,\phi)$ will be labelled by the type of unipotent element u is inside $Z_{G^\vee}(s)^0$, together with the representation ϕ. For example, if u has type $F_4(a_3)$ and s is a two-cycle, then $Z_{G^\vee}(s)$ has type $C_3 \times A_1$, u is labelled by $C_3(42) \times A_1$ (subregular in the first group and regular in the second) and $\widehat{A_{su}} = C_2 \times C_2$. Thus, each representation has a name of the form $[u \in Z_{G^\vee}(s)^0, \phi]$ where ϕ is an irreducible representation of A_{su}.

Let us briefly discuss how the parahoric restrictions are computed. The unipotent representations of $F_4(F)$ fall into 9 unipotent blocks, together with their associated Hecke algebras. There are 7 supercuspidal blocks, one for each cuspidal unipotent representation of $F_4(\mathbb{F}_q)$. The corresponding Hecke algebras $\mathcal{H}(F_4(F), F_4(\mathcal{O}_F), \sigma)$ are all isomorphic to $\mathbb{C}$ so these blocks contain a unique unipotent representation. This information can be read from Lusztig's arithmetic-geometric correspondence tables in [Lus95, pg. 577]. The Langlands parameters of these representations are

$$[F_4(a_3), 1^4],\ [C_3(42), --],\ [B_4(531), \varepsilon],\ [2A_2, \theta],\ [2A_2, \theta^2],\ [A_1A_3, i] \text{ and } [A_1A_3, -i].$$

The parahoric restrictions of these representations are straightforward to compute, for they are irreducible for the hyperspecial parahoric K_0 and vanish for the other ones.

The principal block contains all Iwahori-spherical representations, and the corresponding Hecke algebra $\mathcal{H}(F_4(F), I, \mathbf{1})$ is the affine Hecke algebra of type F_4 with equal parameters. This block contains all representations $\pi(us, \phi)$ such that $H(\mathcal{B}_s^u)^\phi \neq 0$. The parahoric restrictions to the maximal hyperspecial K_0 are computed using Springer theory and the Green functions of $F_4(\mathbb{F}_q)$, which are explicitly known thanks to the work of Beynon-Spaltenstein in Cite BeynonSpaltenstein1984. The results are tabulated at the end of the section. To compute the parahoric restrictions to the other maximal parahoric subgroups, we use (15), which relies on the computation of the groups $W_{J,t}$ and the characters χ_t^J. In many instances, the characters are trivial (for example, if $W_{J,t}$ is a reflection subgroup of W_J); when this is the case, we use Lemma 4.11. The restriction can then be easily obtained from Alvis' tables Insert the reference here.

Finally, there is an intermediate block whose Hecke algebra $\mathcal{H}(F_4(F), B_2, \sigma)$ is the affine Hecke algebra of type C_2 with unequal parameters that we introduced in Example 4.12. All Langlands parameters of the representations of interest in this block are *elliptic*, so it suffices to compute the square integrable representations of $\mathcal{H}(F_4(F), B_2, \sigma)$ to determine the desired parahoric restrictions. This can be achieved using the theory of weight diagrams briefly discussed in Section 4.3 (see also [Ciu20, §9]). The Langlands parameters of these representations are

$$[B_4, -],\ [C_3A_1, -],\ [C_3(42)A_1, -+],\ [B_4(531), r] \text{ and } [A_1A_3, -1].$$

We explicitly discuss the $[A_1A_3, -1]$ case in Example 5.5, and we tabulate the rest of the parahoric restrictions at the end of the section.

Before we end this section with the tables, let us discuss a few examples in detail to illustrate the main ideas involved in the calculations. We will focus on the representations whose unipotent part has type $F_4(a_3)$, since they are the most interesting.

5.1 Examples of parahoric restrictions

The first example is the computation of a parahoric restrictions with respect to the maximal hyperspecial parahoric K_0 using Springer Theory and Alvis' tables.

Example 5.2. Consider the representation $[B_4(531), \mathbf{1}]$, with u of type $F_4(a_3)$ and $s \in \Gamma_u$ a product of two 2-cycles. The centralizer $Z_{G^\vee}(s)$ has type B_4 and $u \in Z_{G^\vee}(s)$ is subregular. Thus, the Springer fibre $\mathcal{B}_s^u$ is 2 dimensional, and by reading the Green functions we obtain

$$H^*(\mathcal{B}_s^u)^{\mathbf{1}} = [2,2] + [3,1] + [4,\cdot] \quad \text{as } W_s\text{-modules.}$$

Since all characters are trivial when $J = J_0$, Alvis' tables yield

$$\mathrm{Ind}_{W_s}^{W}[2,2] + [3,1] + [4,\cdot] = \phi_{9,2} + \phi''_{9,6} + \phi''_{8,3} + \phi_{4,1} + \phi''_{2,4} + \phi_{1,0}.$$

To obtain the parahoric restriction of the representation $[B_4(531), \mathbf{1}]$ with respect to K_0, we twist by ε and by the outer automorphism of W swapping long and short roots, so

$$[B_4(531), \mathbf{1}] \xrightarrow{\mathrm{res}_{\mathrm{un}}^{K_0}} \phi_{9,10} + \phi''_{9,6} + \phi''_{8,9} + \phi_{4,13} + \phi''_{2,16} + \phi_{1,24}.$$

Once the parahoric restrictions for the maximal hyperspecial have been calculated, we can turn our attention towards computing the rest. The first step is to determine when the characters χ_t^J are trivial using Lemma 4.11. The order of the semisimple elements s in the Langlands parameters can be computed easily, while one can show that $Z_{J_1}, Z_{J_2}, Z_{J_3}$ and Z_{J_4} have sizes $2, 2, 3$ and 2, respectively. Whenever the orders of s and J_1 are relatively prime, the characters are trivial, and we can use Lemma 4.11 and Alvis' tables to compute the restriction directly. Let us discuss another interesting example.

Example 5.3. Consider the parahoric restriction of the Iwahori-spherical representation $[A_1A_3, \mathbf{1}]$, where u is of type $F_4(a_3)$ and $s \in \Gamma_u$ is a 4-cycle, with respect to the parahoric subgroup K_2. Note that $W_{J_2} = \langle s_0, s_4, s_3, s_1 \rangle \cong W(A_3) \times W(A_1)$. The semisimple part s can be chosen to be $s = \check{\alpha}_3(-1)\check{\alpha}_4(i)$ so that $W_s = \langle s_4, s_2, s_1, s^0 \rangle \cong W(A_1) \times W(A_3)$. In addition, u is a regular unipotent element in $Z_{G^\vee}(s)$, so $H^*(\mathcal{B}_s^u)^{\mathbf{1}} = \mathbf{1}$ as a W_s-module. Next,

$$W_s \backslash W / W_{J_2} = \{1, w\},$$

where $w \in W$ is explicitly computable with computer algebra programs such as GAP. In addition,

$$W_{J_2,s} = \langle s_4, s_1 \rangle \cong C_2 \times C_2, \quad \text{while} \quad W_{J_2,s^w} = \langle d \rangle \cong C_4,$$

for some $d \in W$ of order 4. Since $W_{J_2,s}$ is a reflection subgroup of W_{J_2}, the character $\chi_s^{J_2}$ is trivial. On the other hand, W_{J_2,s^w} is not a reflection subgroup of W_{J_2}, and $\chi_{s^w}^{J_2} = \chi$ is a primitive character of C_4. The induced representations are then given by

$$\mathrm{Ind}_{W_{J_2,s}}^{W_{J_2}} \mathbf{1} = [[4] + 2[311] + [22] + [211]] \boxtimes [2] \quad \text{and} \quad \mathrm{Ind}_{W_{J_2,s^w}}^{W_{J_2}} \chi = [[31] + [211]] \boxtimes [[2] + [11]].$$

Adding these and twisting by sign gives the required parahoric restriction.

Example 5.4. A more challenging example is the calculation of the parahoric restriction of $[2A_2, \mathbf{1}]$, with u of type $F_4(a_3)$ and $s \in \Gamma_u$ is a 3-cycle, with respect to K_3. This is the only case where the characters $\chi^{J_3}_{s^w}$ may be non-trivial. Note that $W_{J_3} = \langle s_0, s_4, s_2, s_1\rangle \cong W(A_2)\times W(A_2)$. The semisimple part can be taken to be $s = \check{\alpha}_3(\xi_3)\check{\alpha}_4(\xi_3^2)$, where ξ_3 is a primitive third root of unity. In particular, $Z_{G^\vee}(s)$ is a pseudo-Levi subgroup of type $A_2\times A_2$ generated by the affine simple roots $\{\alpha_4, \alpha_3, \alpha_1, -\alpha^0\}$ and $W_s = \langle s_4, s_3, s_1, s^0\rangle$. In addition, the unipotent element u is regular in $Z_{G^\vee}(s)$, so $H^*(\mathcal{B}^u_s)^{\mathbf{1}} = \mathbf{1}$ as W_s-modules. Next,

$$W_s\backslash W/W_{J_3} = \{w_1 = 1, w_2, w_3, w_4, w_5\},$$

where $w_i \in W$ are representatives of double cosets with sizes $324, 324, 36, 36$ and 432, respectively. The subgroups

$$W_{J_3,s^{w_3}} = W_{J_3,s^{w_4}} = W_{J_3} \quad \text{and} \quad W_{J_3,s} \cong W_{J_3,s^{w_2}} \cong C_2\times C_2$$

are all reflections subgroups of W_{J_3}, so the characters $\chi^{J_3}_{s^{w_i}}$ are trivial for $i = 1, 2, 3, 4$. The remaining coset is more interesting since

$$W_{J_3,s^{w_5}} = \langle d\rangle \cong C_3,$$

is no longer a reflection subgroup, and $\chi^{J_3}_{s^{w_4}}$ is a primitive character χ of C_3.

The induced representations are then given by

$$\mathrm{Ind}^{J_3}_{W_{J_3,s}}\mathbf{1} = \mathrm{Ind}^{J_3}_{W_{J_3,s^{w_2}}}\mathbf{1} = [3,3]+[3,21]+[21,3]+[21,21], \quad \mathrm{Ind}^{J_3}_{W_{J_3,s^{w_3}}}\mathbf{1} = \mathrm{Ind}^{J_3}_{W_{J_3,s^{w_4}}}\mathbf{1} = [3,3]$$
$$\text{and} \quad \mathrm{Ind}^{J_3}_{W_{J_3,s^{w_5}}}\chi = [21,3]+[3,21]+[21,21]+[21,1^3]+[1^3,21].$$

Adding these and twisting by sign gives the required parahoric restriction.

Finally, let us discuss the parahoric restrictions of the non-Iwahori-spherical representation $[A_1A_3, -1]$.

Example 5.5. Let V be the representation labelled by $[A_1A_3, -1]$, K a parahoric subgroup of type B_2 and let σ be the unique cuspidal unipotent representation of $B_2(\mathbb{F}_q)$. Since the types of the reductive quotients of K_2 and K_3 ($A_3\times\tilde{A}_1$ and $A_2\times\tilde{A}_2$, respectively) do not contain B_2 as a factor, the parahoric restrictions of V to K_2 and K_3 vanish.

On the other hand, V^σ is a square integrable representation of $\mathcal{H}(F_4(F), B_2, \sigma) \cong \mathcal{H}(K_{J_0}, K, \sigma)\tilde{\otimes}\mathcal{O}[\mathbf{T}]$, whose restriction to $\mathcal{O}[\mathbf{T}]$ is

$$V^\sigma|_{\mathcal{O}[\mathbf{T}]} = V^\sigma(q^{-2}, -q^{-1}) \oplus V^\sigma(-q^{-1}, q^{-2}),$$

both subspaces being one-dimensional. By choosing a basis element in each subspace, $e_{\alpha_1}, e_{\alpha_2} \in \mathcal{O}[\mathbf{T}]$ act on V^σ by $\mathrm{diag}(-q^{-1}, -q)$ and $\mathrm{diag}(-q^{-1}, q^{-2})$. Using the formula for the twisted multiplication in $\mathcal{H}(F_4(F), B_2, \sigma)$, the action of the generators T_1 and T_2 can be computed explicitly, and we obtain

$$T_{s_1} \longrightarrow \frac{1}{q+1}\begin{pmatrix} q^3-1 & q^3+q \\ q^4+1 & q^4-q\end{pmatrix} \quad \text{and} \quad T_{s_2} \longrightarrow \begin{pmatrix} -1 & 0 \\ 0 & -1\end{pmatrix}.$$

Finally, one can trace back through the isomorphisms $\mathcal{H}(F_4(F), B_2, \sigma) \cong \mathcal{H}(K_{J_0}, K, \sigma) \tilde{\otimes} \mathcal{O}[\mathbf{T}]$ to obtain that $T_{\tilde{s}_0}$ acts by

$$T_{\tilde{s}_0} \longrightarrow \frac{1}{q^3+q^2}\begin{pmatrix} -q^6+q^3-q^2-1 & q^5+q^3-q^2-1 \\ -q^7+q^4-q^3+1 & q^6+q^4-q^3+1 \end{pmatrix}.$$

Thus, the action of $\tilde{W}(K,\sigma)$ on $V^{\sigma}_{q=1}$ is given by

$$s_1 \longrightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad s_2 \longrightarrow \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad \tilde{s}_0 \longrightarrow \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

The parahoric restrictions are now obtained by restricting to reflection subgroups. Restriction to the reflection subgroup $\langle s_1, s_2\rangle \cong W(B_2)$ gives the parahoric restriction to K_0, restriction to $\langle s_1, \tilde{s}_0\rangle \cong W(B_2)$ gives the parahoric restriction to K_1 and restriction to $\langle s_2, \tilde{s}_0\rangle \cong W(A_1 \times A_1)$ gives the parahoric restriction to K_4. In each case, there is a bijection between representations of the Weyl group and unipotent representations of $\overline{K}_i$ in the series generated by σ, which we use implicitly. Throughout, we use Lusztig's notation for the unipotent representations. We obtain

- Firstly, $V^{\sigma}|_{\langle s_1,s_2\rangle} = [11,\emptyset] + [\emptyset, 11]$, so $[A_1A_3, -1] \xrightarrow{\mathrm{res}^{K_0}_{\mathrm{un}}} B_2[\varepsilon'] + B_2[\varepsilon]$.
- Secondly, $V^{\sigma}|_{\langle s_1,\tilde{s}_0\rangle} = [1,1]$, so $[A_1A_3, -1] \xrightarrow{\mathrm{res}^{K_1}_{\mathrm{un}}} \left(\begin{smallmatrix} 0 & 1 & 2 & 4 \\ & & 1 & \end{smallmatrix}\right) = \theta_3$.
- Finally, $V^{\sigma}|_{\langle s_2,\tilde{s}_0\rangle} = [11] \boxtimes [2] + [11] \boxtimes [11]$, so $[A_1A_3, -1] \xrightarrow{\mathrm{res}^{K_4}_{\mathrm{un}}} \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & & 1 & \end{smallmatrix}\right) \boxtimes \mathbf{1} + \left(\begin{smallmatrix} 0 & 1 & 2 & 3 \\ & & 1 & \end{smallmatrix}\right) \boxtimes \mathrm{St}$.

5.2 Parahoric restriction tables for F_4

Let us clarify some of the notation used in the tables below. The labelling on the representations of F_4 used by Reeder is discussed above. The unipotent representations of finite reductive groups of Lie type are labelled as follows. For groups of type A_n, these are parametrized by partitions λ of $n+1$. For groups of type B_n/C_n, the principal series representations are parametrized by double partitions $[\lambda, \mu]$ of n. The classification of non-principal series is explained in section careful with this reference. To understand these tables, it is enough to know that groups of type B_3/C_3 have two non-principal series representations θ_1 and θ_2, while groups of type B_4/C_4 have five non-principal series representations $\theta_1, \dots, \theta_5$. These all come from the unipotent cuspidal representation of their Levi subgroup of type B_2/C_2.

For every parahoric subgroup K_i, we present two tables. The first one is the table of parahoric restrictions for all Iwahori-spherical representations of $F_4(F)$ of interest. The second table contains the parahoric restrictions of all virtual characters $\Pi(u, s, h)$. The representations of $\overline{K}_i$ are organized in terms of the families. Firstly, we write the characters in trivial families; then we write each family, with the special character always coming first.

From these tables, the statement of Theorem 5.1 can be easily verified using Lusztig's nonabelian transform. For reference, we write here the virtual characters $\Pi(u, s, h)$ in terms of representations of $F_4(F)$ when $u \in F_4(a_3)$. In the notation below, $\tau = g_2 \cdot g_2'$ is another 2-cycle conjugate to g_2 in $A_u = S_4$ but not in $A_{ug_2} = C_2 \times C_2$ and, similarly, γ is an element conjugate to g_2' in $A_u = S_4$ but not in $A_{ug_2'} = D_8$. The analogous expressions for other unipotent classes are much simpler.

$$\Pi(u,1,1) = [F_4(a_3),4] + 3[F_4(a_3),31] + 3[F_4(a_3),211] + [F_4(a_3),1^4] + 2[F_4(a_3),22];$$
$$\Pi(u,1,g_2) = [F_4(a_3),4] + [F_4(a_3),31] - [F_4(a_3),211] - [F_4(a_3),1^4];$$
$$\Pi(u,1,g_2') = [F_4(a_3),4] - [F_4(a_3),31] - [F_4(a_3),211] + [F_4(a_3),1^4] + 2[F_4(a_3),22];$$
$$\Pi(u,1,g_3) = [F_4(a_3),4] + [F_4(a_3),1^4] - [F_4(a_3),22];$$
$$\Pi(u,1,g_4) = [F_4(a_3),4] - [F_4(a_3),31] + [F_4(a_3),211] - [F_4(a_3),1^4];$$

$$\Pi(u,g_2,1) = [C(42)A_1,++] + [C(42)A_1,+-] + [C(42)A_1,-+] + [C(42)A_1,--];$$
$$\Pi(u,g_2,g_2) = [C(42)A_1,++] - [C(42)A_1,+-] + [C(42)A_1,-+] - [C(42)A_1,--];$$
$$\Pi(u,g_2,\tau) = [C(42)A_1,++] + [C(42)A_1,+-] - [C(42)A_1,-+] - [C(42)A_1,--];$$
$$\Pi(u,g_2,g_2') = [C(42)A_1,++] - [C(42)A_1,+-] - [C(42)A_1,-+] + [C(42)A_1,--];$$

$$\Pi(u, g_2', 1) = [B_4(531), \mathbf{1}] + 2[B_4(531), r] + [B_4(531), \varepsilon'] + [B_4(531), \varepsilon''] + [B_4(531), \varepsilon];$$

$$\Pi(u, g_2', g_2') = [B_4(531), \mathbf{1}] - [B_4(531), \varepsilon'] + [B_4(531), \varepsilon''] - [B_4(531), \varepsilon];$$

$$\Pi(u, g_2', g_2'') = [B_4(531), \mathbf{1}] - 2[B_4(531), r] + [B_4(531), \varepsilon'] + [B_4(531), \varepsilon''] + [B_4(531), \varepsilon];$$

$$\Pi(u, g_2', g_4) = [B_4(531), \mathbf{1}] - [B_4(531), \varepsilon'] - [B_4(531), \varepsilon''] + [B_4(531), \varepsilon];$$

$$\Pi(u, g_2', \gamma) = [B_4(531), \mathbf{1}] + [B_4(531), \varepsilon'] - [B_4(531), \varepsilon''] - [B_4(531), \varepsilon];$$

$$\Pi(u, g_3, 1) = [2A_2, \mathbf{1}] + [2A_2, \theta] + [2A_2, \theta^2];$$

$$\Pi(u, g_3, g_3) = [2A_2, \mathbf{1}] + \theta^2[2A_2, \theta] + \theta[2A_2, \theta^2];$$

$$\Pi(u, g_3, g_3^{-1}) = [2A_2, \mathbf{1}] + \theta[2A_2, \theta] + \theta^2[2A_2, \theta^2];$$

$$\Pi(u, g_4, 1) = [A_3A_1, 1] + [A_3A_1, i] + [A_3A_1, -1] + [A_3A_1, -i];$$

$$\Pi(u, g_4, g_4) = [A_3A_1, 1] - i[A_3A_1, i] - [A_3A_1, -1] - i[A_3A_1, -i];$$

$$\Pi(u, g_4, g_4') = [A_3A_1, 1] - [A_3A_1, i] + [A_3A_1, -1] - [A_3A_1, -i];$$

$$\Pi(u, g_4, g_4^{-1}) = [A_3A_1, 1] + i[A_3A_1, i] - [A_3A_1, -1] - i[A_3A_1, -i];$$

Iwahori-spherical B_4-types

	$[4,\cdot]$	$[31,\cdot]$	$[22,\cdot]$	$[21^2,\cdot]$	$[1^4,\cdot]$	$[3,1]$	$[21,1]$	$[1^3,1]$	$[2,2]$	$[2,11]$	$[11,2]$	$[11,11]$	$[1,3]$	$[1,21]$	$[1,1^3]$	$[\cdot,4]$	$[\cdot,31]$	$[\cdot,22]$	$[\cdot,21^2]$	$[\cdot,1^4]$
$[F_4,1]$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
$[F_4(a_1),+]$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
$[F_4(a_1),-]$	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
$[B_4,+]$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
$[F_4(a_2),+]$	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1
$[F_4(a_2),-]$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
$[C_3A_1,+]$	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	1
$[B_4(711),+]$	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	1	0
$[B_4(711),-]$	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	1
$[F_4(a_3),4]$	0	0	0	0	0	0	0	1	0	1	1	1	0	1	2	0	0	0	1	1
$[F_4(a_3),31]$	0	0	0	1	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0	1
$[F_4(a_3),22]$	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0
$[F_4(a_3),211]$	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
$[C_3(42)A_1,++]$	0	0	0	1	0	0	0	1	0	1	1	1	0	1	2	0	0	0	2	1
$[C_3(42)A_1,+-]$	0	0	0	0	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	1
$[B_4(531),1]$	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1	0	1	0	2	0
$[B_4(531),\varepsilon']$	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
$[B_4(531),\varepsilon'']$	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
$[2A_2,1]$	0	0	0	0	0	0	0	1	0	0	1	1	0	1	1	0	0	0	1	1
$[A_1A_3,1]$	0	0	0	0	0	0	0	0	0	0	1	0	0	1	1	0	1	0	1	0

Table 3: Parahoric restriction of unipotent $F_4(F)$-representations with respect to K_1

	$[4,\cdot]$	$[21,1]$	$[11,2]$	$[1,21]$	$[\cdot,1^4]$	$[1,1^3]$	$[1^4,\cdot]$	$[\cdot,21^2]$	θ_1	$[11,11]$	$[1^3,1]$	$[\cdot,22]$	θ_2	$[2,11]$	$[21^2,\cdot]$	$[\cdot,31]$	θ_3	$[2,2]$	$[22,\cdot]$	$[1,3]$	θ_4
$\Pi(u,1,1)$	0	0	1	1	4	7	6	1	0	4	4	0	0	3	3	0	0	0	0	0	0
$\Pi(u,1,g_2)$	0	0	1	1	2	3	0	1	0	2	2	0	0	1	1	0	0	0	0	0	0
$\Pi(u,1,g_2')$	0	0	1	1	0	3	-2	1	0	0	0	0	0	3	-1	0	0	0	0	0	0
$\Pi(u,1,g_3)$	0	0	1	1	1	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0
$\Pi(u,1,g_4)$	0	0	1	1	0	1	0	1	0	0	0	0	0	1	-1	0	0	0	0	0	0
$\Pi(u,g_2,1)$	0	0	1	1	2	2	1	2	1	2	2	0	0	1	1	0	0	0	0	0	0
$\Pi(u,g_2,g_2)$	0	0	1	1	2	2	1	2	-1	2	2	0	0	1	1	0	0	0	0	0	0
$\Pi(u,g_2,\tau)$	0	0	1	1	0	2	-1	2	1	0	0	0	0	1	1	0	0	0	0	0	0
$\Pi(u,g_2,g_2')$	0	0	1	1	0	2	-1	2	-1	0	0	0	0	1	1	0	0	0	0	0	0
$\Pi(u,g_2',1)$	0	0	1	1	0	1	0	3	2	0	0	0	0	1	1	2	2	0	0	0	0
$\Pi(u,g_2',g_2)$	0	0	1	1	0	1	0	3	0	0	0	0	0	1	1	0	0	0	0	0	0
$\Pi(u,g_2',g_2')$	0	0	1	1	0	1	0	3	-2	0	0	0	0	1	1	2	-2	0	0	0	0
$\Pi(u,g_2',\gamma)$	0	0	1	1	0	1	0	1	0	0	0	0	0	1	-1	2	0	0	0	0	0
$\Pi(u,g_2',g_4)$	0	0	1	1	0	1	0	1	0	0	0	0	0	1	-1	0	0	0	0	0	0
$\Pi(u,g_3,h)$	0	0	1	1	1	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0
$\Pi(u,g_4,1/g_2')$	0	0	1	1	0	1	0	1	0	0	0	0	0	0	0	1	1	0	0	0	0
$\Pi(u,g_4,g_4^{\pm 1})$	0	0	1	1	0	1	0	1	0	0	0	0	0	0	0	1	-1	0	0	0	0

Table 4: Parahoric restrictions of the virtual characters $\Pi(u,s,h)$ where $u \in F_4(a_3)$ with respect to K_1

Iwahori-spherical $A_3 \times A_1$-types

	$[1^4]$	$[211]$	$[22]$	$[31]$	$[4]$
$[F_4, 1]$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_1), +]$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_1), -]$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4, +]$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_2), +]$	$1,\bar{2}$	$1,\bar{2}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_2), -]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$
$[C_3A_1, +]$	$1,\bar{2}$	$1,\bar{2}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(711), +]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(711), -]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_3), 4]$	$2,\bar{3}$	$3,\bar{5}$	$1,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$[F_4(a_3), 31]$	$3,\bar{2}$	$2,\bar{2}$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_3), 22]$	$1,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$
$[F_4(a_3), 211]$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[C_3(42)A_1, ++]$	$2,\bar{3}$	$4,\bar{6}$	$1,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$[C_3(42)A_1, +-]$	$2,\bar{2}$	$1,\bar{1}$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(531), 1]$	$0,\bar{1}$	$2,\bar{4}$	$0,\bar{1}$	$1,\bar{3}$	$0,\bar{0}$
$[B_4(531), \varepsilon'']$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(531), \varepsilon']$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$
$[2A_2, 1]$	$1,\bar{3}$	$2,\bar{4}$	$1,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$
$[A_1A_3, 1]$	$0,\bar{1}$	$1,\bar{3}$	$0,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$

Table 5: Parahoric restriction of $F_4(F)$-representations labelled by the distinguished unipotent $F_4(a_3)$ with respect to K_2

	$[1^4]$	$[211]$	$[22]$	$[31]$	$[4]$
$\Pi(u, 1, 1)$	$16,\bar{9}$	$11,\overline{13}$	$4,\bar{1}$	$1,\bar{4}$	$0,\bar{0}$
$\Pi(u, 1, g_2)$	$4,\bar{5}$	$5,\bar{7}$	$2,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, 1, g_2')$	$0,\bar{1}$	$3,\bar{5}$	$0,\bar{1}$	$1,\bar{4}$	$0,\bar{0}$
$\Pi(u, 1, g_3)$	$1,\bar{3}$	$2,\bar{4}$	$1,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$
$\Pi(u, 1, g_4)$	$0,\bar{1}$	$1,\bar{3}$	$0,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, g_2, 1)$	$4,\bar{5}$	$5,\bar{7}$	$2,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, g_2, g_2)$	$4,\bar{5}$	$5,\bar{7}$	$2,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, g_2, \tau)$	$0,\bar{1}$	$3,\bar{5}$	$0,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, g_2, g_2')$	$0,\bar{1}$	$3,\bar{5}$	$0,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, g_2', 1)$	$0,\bar{1}$	$3,\bar{5}$	$0,\bar{1}$	$1,\bar{4}$	$0,\bar{0}$
$\Pi(u, g_2', g_2)$	$0,\bar{1}$	$3,\bar{5}$	$0,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, g_2', \gamma)$	$0,\bar{1}$	$1,\bar{3}$	$0,\bar{1}$	$1,\bar{4}$	$0,\bar{0}$
$\Pi(u, g_2', g_2')$	$0,\bar{1}$	$3,\bar{5}$	$0,\bar{1}$	$1,\bar{4}$	$0,\bar{0}$
$\Pi(u, g_2', g_4)$	$0,\bar{1}$	$1,\bar{3}$	$0,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$
$\Pi(u, g_3, h)$	$1,\bar{3}$	$2,\bar{4}$	$1,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$
$\Pi(u, g_4, h)$	$0,\bar{1}$	$1,\bar{3}$	$0,\bar{1}$	$1,\bar{2}$	$0,\bar{0}$

Table 6: Parahoric restrictions of the virtual characters $\Pi(u, s, h)$ where $u \in F_4(a_3)$ with respect to K_2

Iwahori-spherical $A_2 \times A_2$-types

	$[1^3, 1^3]$	$[1^3, 21]$	$[1^3, 3]$	$[21, 1^3]$	$[21, 21]$	$[21, 3]$	$[3, 1^3]$	$[3, 21]$	$[3, 3]$
$[F_4, 1]$	1	0	0	0	0	0	0	0	0
$[F_4(a_1), +]$	1	1	0	1	0	0	0	0	0
$[F_4(a_1), -]$	0	1	0	0	0	0	0	0	0
$[B_4, +]$	1	0	0	1	0	0	0	0	0
$[F_4(a_2), +]$	2	2	0	2	1	0	0	0	0
$[F_4(a_2), -]$	0	0	0	1	0	0	0	0	0
$[C_3A_1, +]$	2	2	0	1	1	0	0	0	0
$[B_4(711), +]$	0	0	0	0	0	0	0	0	0
$[B_4(711), -]$	0	0	0	0	1	0	0	0	0
$[F_4(a_3), 4]$	4	4	1	4	4	1	1	1	0
$[F_4(a_3), 31]$	1	3	2	0	2	1	0	0	0
$[F_4(a_3), 22]$	0	1	1	1	1	0	1	0	0
$[F_4(a_3), 211]$	0	0	1	0	0	0	0	0	0
$[C_3(42)A_1, ++]$	4	5	1	4	5	1	1	1	0
$[C_3(42)A_1, +-]$	1	2	1	0	1	1	0	0	0
$[B_4(531), 1]$	3	2	0	5	3	0	2	1	0
$[B_4(531), \varepsilon'']$	0	1	0	0	1	0	0	0	0
$[B_4(531), \varepsilon']$	0	0	0	1	0	0	1	0	0
$[2A_2, 1]$	4	3	0	3	3	1	0	1	0
$[A_1A_3, 1]$	3	1	0	4	2	0	1	1	0

Table 7: Parahoric restriction of $F_4(F)$-representations labelled by the distinguished unipotent $F_4(a_3)$ with respect to K_3

	$[1^3,1^3]$	$[1^3,21]$	$[1^3,3]$	$[21,1^3]$	$[21,21]$	$[21,3]$	$[3,1^3]$	$[3,21]$	$[3,3]$
$\Pi(u,1,1)$	7	15	12	6	12	4	3	1	0
$\Pi(u,1,g_2)$	5	7	2	4	6	2	1	1	0
$\Pi(u,1,g_2')$	3	3	0	6	4	0	3	1	0
$\Pi(u,1,g_3)$	4	3	0	3	3	1	0	1	0
$\Pi(u,1,g_4)$	3	1	0	4	2	0	1	1	0
$\Pi(u,g_2,1)$	5	7	2	4	6	2	1	1	0
$\Pi(u,g_2,g_2)$	5	7	2	4	6	2	1	1	0
$\Pi(u,g_2,\tau)$	3	3	0	4	4	0	1	1	0
$\Pi(u,g_2,g_2')$	3	3	0	4	4	0	1	1	0
$\Pi(u,g_2',1)$	3	3	0	6	4	0	3	1	0
$\Pi(u,g_2',g_2)$	3	3	0	4	4	0	1	1	0
$\Pi(u,g_2',\gamma)$	3	1	0	6	2	0	3	1	0
$\Pi(u,g_2',g_2')$	3	3	0	6	4	0	3	1	0
$\Pi(u,g_2',g_4)$	3	1	0	4	2	0	1	1	0
$\Pi(u,g_3,h)$	4	3	0	3	3	1	0	1	0
$\Pi(u,g_4,h)$	3	1	0	4	2	0	1	1	0

Table 8: Parahoric restrictions of the virtual characters $\Pi(u,s,h)$ where $u \in F_4(a_3)$ with respect to K_3

Iwahori-spherical $A_1 \times C_3$-types

	$[\cdot,1^3]$	$[1^3,\cdot]$	$[11,1]$	$[1,11]$	$[21,\cdot]$	$[\cdot,21]$	$[2,1]$	$[1,2]$	$[\cdot,3]$	$[3,\cdot]$
$[F_4,1]$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_1),+]$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_1),-]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4,+]$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_2),+]$	$1,\bar{2}$	$0,\bar{0}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_2),-]$	$0,\bar{1}$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[C_3A_1,+]$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(711),+]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(711),-]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_3),4]$	$2,\bar{2}$	$0,\bar{1}$	$1,\bar{2}$	$2,\bar{3}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_3),31]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$1,\bar{2}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$
$[F_4(a_3),22]$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[F_4(a_3),211]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$
$[C_3(42)A_1,++]$	$2,\bar{2}$	$0,\bar{1}$	$1,\bar{2}$	$2,\bar{3}$	$0,\bar{1}$	$2,\bar{2}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$
$[C_3(42)A_1,+-]$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{1}$	$0,\bar{0}$
$[B_4(531),1]$	$3,\bar{2}$	$1,\bar{2}$	$1,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(531),\varepsilon'']$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[B_4(531),\varepsilon']$	$1,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$[2A_2,1]$	$1,\bar{2}$	$0,\bar{1}$	$1,\bar{2}$	$1,\bar{2}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$
$[A_1A_3,1]$	$2,\bar{2}$	$1,\bar{2}$	$1,\bar{2}$	$1,\bar{1}$	$0,\bar{0}$	$1,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$

Table 9: Parahoric restriction of $F_4(F)$-representations labelled by the distinguished unipotent $F_4(a_3)$ with respect to K_4

	$[3,\cdot]$	$[1,2]$	$[11,1]$	$[\cdot,1^3]$	$[1,11]$	$[1^3,\cdot]$	$[\cdot,21]$	θ_1	$[2,1]$	$[21,\cdot]$	$[\cdot,3]$	θ_2
$\Pi(u,1,1)$	$0,\bar{0}$	$4,\bar{4}$	$1,\bar{2}$	$4,\bar{2}$	$4,\bar{8}$	$0,\bar{1}$	$4,\bar{7}$	$0,\bar{0}$	$0,\bar{6}$	$0,\bar{0}$	$0,\bar{6}$	$0,\bar{0}$
$\Pi(u,1,g_2)$	$0,\bar{0}$	$2,\bar{2}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{4}$	$0,\bar{1}$	$2,\bar{3}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,1,g_2')$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$4,\bar{4}$	$0,\bar{1}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,-\bar{2}$	$0,\bar{0}$
$\Pi(u,1,g_3)$	$0,\bar{0}$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{2}$	$1,\bar{2}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,1,g_4)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,g_2,1)$	$0,\bar{0}$	$2,\bar{2}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{4}$	$0,\bar{1}$	$2,\bar{3}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{1}$
$\Pi(u,g_2,g_2)$	$0,\bar{0}$	$2,\bar{2}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{4}$	$0,\bar{1}$	$2,\bar{3}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{1}$	$0,-\bar{1}$
$\Pi(u,g_2,\tau)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$2,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{-1}$	$0,\bar{1}$
$\Pi(u,g_2,g_2')$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$2,\bar{1}$	$0,\bar{0}$	$0,\bar{1}$	$0,\bar{1}$	$0,\bar{-1}$	$0,\bar{-1}$
$\Pi(u,g_2',1)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$2,\bar{2}$	$2,\bar{3}$	$2,\bar{1}$	$2,\bar{2}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,\bar{2}$
$\Pi(u,g_2',g_2)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$2,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,g_2',\gamma)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$2,\bar{2}$	$2,\bar{3}$	$0,\bar{-1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,g_2',g_2')$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$4,\bar{2}$	$2,\bar{2}$	$2,\bar{3}$	$2,\bar{1}$	$-2,-\bar{2}$	$0,\bar{0}$	$0,\bar{2}$	$0,\bar{0}$	$0,-\bar{2}$
$\Pi(u,g_2',g_4)$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$2,\bar{2}$	$0,\bar{1}$	$0,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,g_3,h)$	$0,\bar{0}$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{2}$	$1,\bar{2}$	$0,\bar{1}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,g_4,1/g_2')$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{0}$	$1,\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$
$\Pi(u,g_4,g_4^{\pm 1})$	$0,\bar{0}$	$0,\bar{0}$	$1,\bar{2}$	$2,\bar{2}$	$1,\bar{1}$	$1,\bar{2}$	$1,\bar{0}$	$-1,-\bar{1}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$	$0,\bar{0}$

Table 10: Parahoric restrictions of the virtual characters $\Pi(u,s,h)$ where $u \in F_4(a_3)$ with respect to K_4

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